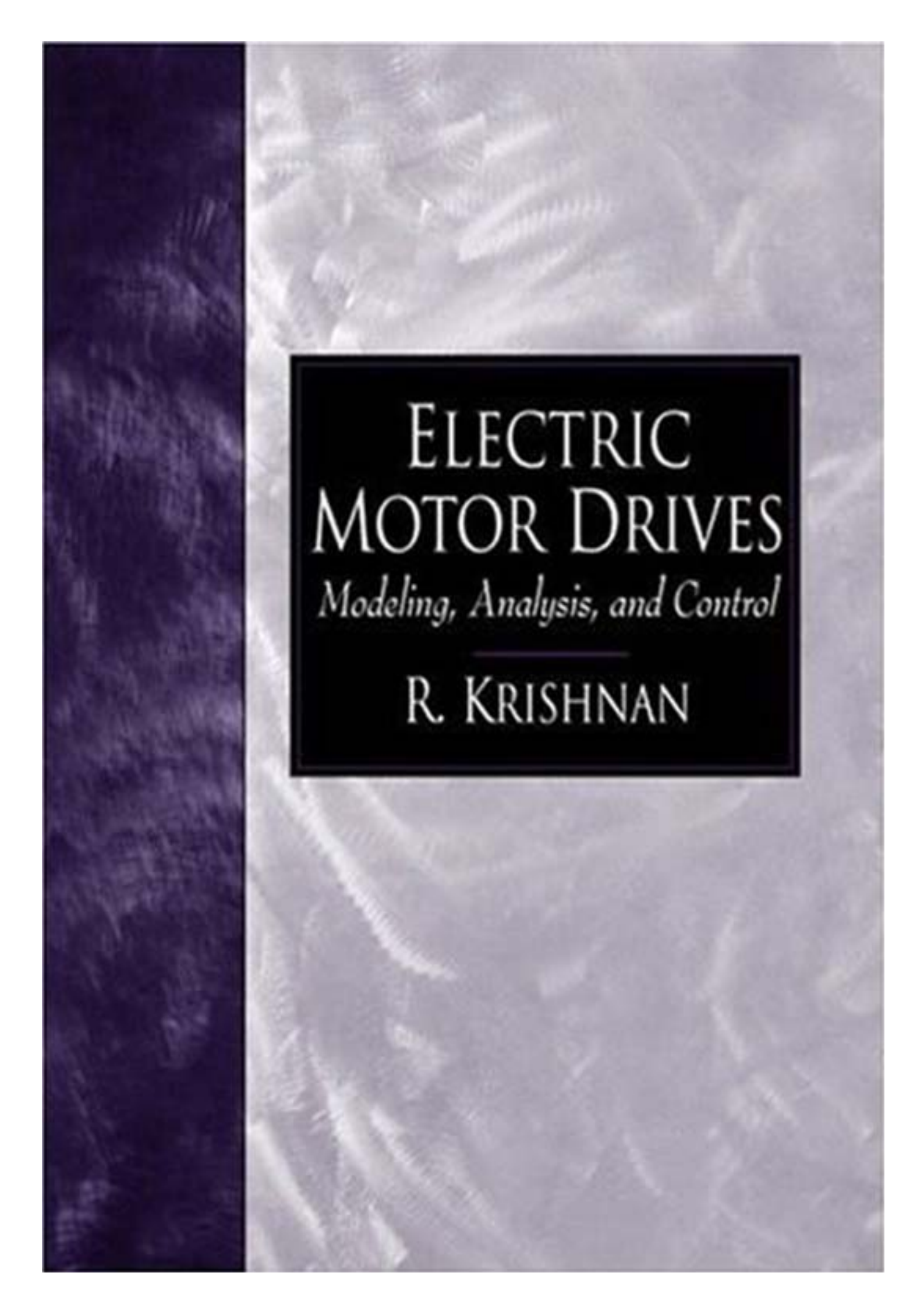




ELECTRIC MOTOR DRIVES

Modeling, Analysis, and Control

R. KRISHNAN

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In fond memory of:

*My grandfather, **Mr. A. Duraiswami Mudaliar**, an Indian Freedom Fighter,*

*My grandmother, **Mrs. D. Rajammal**,*

and

*My father-in-law, **Prof. S. K. Ekambaram**, B.Sc. (Hons.), M.A. (Cantab).*

Preface

Electronic control of machines is a course taught *as an elective at senior level and as an introductory course for the graduate-level course on motor drives*. Many wonderful books address individually the senior-level or graduate-level requirements, mostly from the practitioner's point of view, but some are not ideally suited for classroom teaching in American universities. This book is the result of teaching materials developed over a period of twelve years at Virginia Tech. Parts of the material have been used extensively in seminars both in the United States and abroad.

The area of electric motor drives is a dependent discipline. It is an applied and multidisciplinary subject comprising electronics, machines, control, processors/computers, software, electromagnetics, sensors, power systems, and engineering applications. It is not possible to cover all aspects relevant to motor drives in one text. Therefore, this book addresses mainly the *system-level modeling analysis, design and integration of motor drives*. In this regard, knowledge of electrical machines, power converters, and linear control systems is assumed at the junior level. The modeling and analysis of electrical machines and drive systems is systematically derived from first principles. The control algorithms are developed, and their implementations with simulation results are given wherever appropriate.

The book consists of nine chapters. Their contents are briefly described here.

Chapter 1 contains the introduction and discusses the motor-drive applications, the status of power devices, classes of electrical machines, power converters, controllers, and mechanical systems.

Chapter 2 is on dc machines, their principle of operation, the steady-state and dynamic modeling, block-diagram development, and measurement of motor parameters.

Chapter 3 describes phase-controlled dc motor drives for variable-speed operation. The principle of dc machine speed control is developed, and four-quadrant operation is introduced. The electrical requirements for four-quadrant operation are derived. Realization of these voltage/current requirements for four-quadrant operation with phase-controlled converters is studied. In that

process, the operation and control of phase-controlled converters are developed. The closed-loop speed-controlled dc motor drive system is considered for analysis and controller synthesis. The synthesis of current and speed controllers is developed. The dynamic simulation procedure is derived for the motor drive system from the subsystem differential equations and functional relationships. The same procedure is adopted throughout the book. Impact of harmonics both on the utility and on the machine is analyzed. Supply-side harmonics can cause resonance in the case of interaction with the power systems, as is illustrated with an example, and machine-side harmonics cause increased resistive losses, resulting in derating of the machine. Some application considerations for the drive are given. An application of the drive is described. A more or less similar approach is taken for all other drives in this book.

Chopper-controlled dc motors are described in Chapter 4. The principle of operation of a four-quadrant chopper, realization of dc input supply, regeneration, modeling of a chopper, the closed-loop speed-controlled drive and its current- and speed-controller synthesis, harmonics, and their impact on electromagnetic torque, losses, and derating are developed and presented.

The principle of operation of induction machines and their steady-state and dynamic modeling are presented in Chapter 5. The concept of space-phaser modeling is also introduced, to enable readers to follow literature mainly from Germany. A number of illustrative examples are included.

The principle of speed control of induction motors is introduced in Chapter 6. The rest of the chapter is devoted to the stator-phase control and slip-energy-recovery control of induction motors. Only steady-state aspects are covered. Their dynamic analysis is left to the interested reader. Emphasis is placed on efficiency, energy savings, speed-control range, harmonics, and application for these drives.

Variable-frequency control of induction machines with both variable voltage and variable current is introduced in Chapter 7. The realization of variable voltage and variable frequency with two-stage controllable converters and single-stage pulse-width-modulated (PWM) inverters is introduced. Reducing harmonics with multiple inverters or with one PWM inverter is discussed and is illustrated with examples. Steady-state analysis using fundamental and harmonic equivalent circuits, direct steady-state evaluation, and the use of a dynamic model with boundary-matching conditions is systematically developed in this chapter. Various control strategies for variable-voltage, variable-frequency drives are explained. They are V/Hz, constant-slip-speed, and constant-air gap-flux controls. The limitations and merits of each control scheme and relevant modeling to evaluate their dynamic performance are developed. Effects of harmonics on the machine losses, with the resultant derating and torque pulsation due to six-step voltage input to the machine, are quantified. Torque pulsations are calculated by using harmonic equivalent circuits of the induction motor fed from voltage-source inverters. The concept of current source is introduced by using a PWM inverter with voltage-source and current-feedback control. A two-stage current-source inverter drive with thyristor converter front end and autosequentially commutated inverter is considered for both steady-state and dynamic performance evaluation. The design aspects of the current-source drive control are considered in adequate detail.

The high-performance induction motor drive is considered from the control point of view in Chapter 8. The principle of vector control and design and its various implementations, their strengths and weaknesses, impact of parameter sensitivity, parameter-compensation methods, flux-weakening operation, and the design of a speed controller are explained with detailed algorithms and illustrated with dynamic simulation results and example problems. Tuning of the vector controller and position-sensorless operation are not dealt in detail. Interested readers will be helped by the cited references.

Chapter 9 deals with permanent magnet (PM) synchronous and brushless dc motor drives. The salient differences between these two types of motors are derived. Vector control and various control strategies within the scope of vector control, such as constant-torque-angle control, unity-power-factor control, constant-mutual-flux-linkages control, and maximum-torque-per-unit-current control are derived from the dynamic and steady-state equations of the PM synchronous motor. Two types of flux-weakening operation and their implementations, speed-controller design, position-sensorless control, and parameter sensitivity and its compensation are developed with control algorithms. Ample dynamic simulation results are included to enhance the understanding of the PM synchronous motor drive operation. Similar coverage is carried out for the PM brushless dc motor drive. In addition, an analytical method to evaluate torque pulsation and various unipolar/half-wave inverter topologies is included for low-cost but high-reliability motor drive systems.

A list of symbols has been given to enable readers who skip some sections to follow the text they are interested in. The material in this book is recommended for two semesters. In the author's experience, the subject matter that can be covered for each semester is given only to serve as a guideline. Depending on the strength of the program in each school, course instructors can flexibly choose the material from the book for their lectures. The introductory course that can be taught at senior elective or at graduate level includes the following:

Chapters 1, 2, 3, and 4, Chapter 5 (covering only the steady-state operation and modeling of induction motors), Chapter 6, and Chapter 7 (excluding the dynamic performance evaluation or parts that use dynamic model of the induction motor).

The advanced graduate course can consist of the following:

Dynamic modeling of induction machines from Chapter 5, voltage- and current-source drive dynamic performance from Chapter 7, and Chapters 8 and 9.

The author's many graduate and undergraduate students have enriched this book in the course of its development. Some of them deserve my special thanks and gratitude. They are Mr. Praveen Vijayraghavan, Dr. Shiyong Lee, Dr. A. S. Bharadwaj, Dr. Byeong-Seok Lee and Dr. Ramin Monajemy. Without their crucial help, this endeavor would still be in the manuscript stage. The Chapter 5 development draws heavily from the work of one of my doctoral supervisors, Dr. J. F. Lindsay's material taught in 1980; the portion on the space-phasor model is from Prof. Dr. Ing-J. Holtz's research work. I have been fortunate to have Drs. J. F. Lindsay and V. R. Stefanovic as my doctoral supervisors, and a long-time professional and personal

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My editor at Prentice-Hall, Eric Frank smoothed all the glitches and provided advice on many aspects of the book. Lakshmi Balasubramanian, production editor, interfaced pleasantly during copy editing, proofreading and production. Brian Baker/WriteWith Inc. copy edited in a very short time. Robert Lentz took care of the proofreading. Laserwords provided the graphic work. All of them made the process of bringing the book to reality with such ease, pleasantness and clockwork schedule. I am grateful to this incredible team for working with me.

But for my wife, Vijaya's, encouragement and help, this book would not have been possible. To her, I owe the most.

R. KRISHNAN

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Symbols

A	State-transition matrix
a	Turns ratio
a_n	Product of L_{sn} and σ
B_1, B_2	Bearing friction coefficients, N-m/(rad/sec)
B_l	Load constant
B_n	Normalized friction coefficient, p.u.
B_t	Total friction coefficient, N-m/(rad/sec)
C	Capacitance of the filter, F
c_p	Winding pitch factor
c_d	Winding distribution factor
C_f	Filter capacitance, F
C_o	Value of the C-dump capacitor, F
d	Duty cycle
d_c	Critical duty cycle
e	Instantaneous induced emf, V
E	Steady-state induced emf and also used as C-dump capacitor voltage, V
E_1	RMS air gap induced emf in AC machines, V
E_2	RMS rotor-induced emf in AC machines, V
e_{as}	Induced emf in phase a (instantaneous), V
E_{as}	Steady-state rms phase a induced emf, V
e_{bs}	Induced emf in b phase (instantaneous), V
e_{cs}	Induced emf in c phase (instantaneous), V
e_n	Instantaneous normalized induced emf, p.u.
E_n	Normalized steady-state induced emf, p.u.
E_p	Peak stator phase voltage in PM brushless DC machine, V
F, F^*	Modified reactive power and its reference, VAR
$f_{as}(\theta_r)$	Position-dependent function in the induced emf in a phase
$f_{bs}(\theta_r)$	Position-dependent function in the induced emf in b phase
f_c	Control frequency, and PWM carrier frequency, Hz
$f_{cs}(\theta_r)$	Position-dependent function in the induced emf in c phase
f_n	Natural frequency of the power system, Hz
f_o	System frequency, Hz
f_s	Utility supply frequency or stator supply frequency, Hz
f_s^*	Stator frequency command, Hz
f_{sn}	Normalized stator frequency, p.u.
$G_{wl}(s)$	Speed-to-load-torque transfer function
$G_{wv}(s)$	Speed-to-applied-voltage transfer function
h	Average duty cycle of phase switches in half-wave converter topologies in Ch. 9
h	Harmonic number in Ch. 7
H	Inertial constant in normalized unit, p.u.
H_c	Gain of the current transducer, V/A
H_f	Field-current-transducer gain, V/A
H_w	Gain of the speed filter, V/(rad/sec)
i_0	Zero-sequence current, A
I_6	Sixth-harmonic armature current, A
I_{6np}	Normalized peak sixth-harmonic armature current, p.u.
i_a	DC machine armature current, A
i_a^*	Armature current reference, A
I_{a0}	Steady-state minimum armature current, A
i_{a1}	First-harmonic armature current (instantaneous), A
I_1	First-harmonic armature current (steady state), A

I_{a1}	Steady-state maximum armature current, A
i_{abc}	abc phase current vector
I_{ai}	Initial armature current, A
I_{a11}	Fundamental system input phase current, A
I_{an}	Normalized armature current in dc machine, p.u.
i_{an}^*	Normalized armature current reference, p.u.
I_{an}	Normalized phase a stator current, p.u.
i_{ar}	Rated armature current in dc machine, A
i_{as}	Instantaneous stator phase a current, A
i_{asn}	Normalized phase a stator current (instantaneous), p.u.
I_{avn}	Normalized average armature current, A
I_b	Base current in 2-phase system, A
I_b	Base current, A
I_{b3}	Base current in 3-phase system, A
i_{bs}, i_{cs}	Instantaneous b and c stator phase currents, A
$i_{as}^*, i_{bs}^*, i_{cs}^*$	a, b, c phase-current commands, A
I_c	Core-loss current per phase, A
I_{cn}	n^{th} -harmonic capacitor current, A
I_d	Average diode current, A
i_{dc}	Instantaneous dc link current or inverter input current, A
I_{dc}	Steady-state dc link current, A
I_{dcn}	Normalized dc link current, p.u.
i_{dr}^*, i_{qr}^*	Conjugates of i_{dr} and i_{qr}
i_{drr}, i_{qrr}	Fictitious rotor currents in d and q axis, A
i_{ds}, i_{qs}	d and q axis stator currents, A
I_f	Instantaneous field current in dc machines or flux-producing component of the stator-current phasor in ac machines, A
I_f	Steady-state field current or flux-producing component of the stator-current phasor in ac machines, A
i_f^*	Field-current reference, A
I_{fr}, I_{frn}	Rated values of I_f in SI and normalized units
I_m	Magnetizing current per phase, A
I_{m1}	Fundamental magnetizing current, A
I_n	n^{th} -harmonic current, A
I_o	No-load RMS phase current, A
i_{os}, i_{or}	Stator and rotor zero-sequence currents, A
I_p	Peak stator-phase current in PM brushless dc machine, A
I_{ph}	Fundamental phase current, A
I_{ps}	Peak stator-phase current in PM synchronous machine, A
i_{qdo}	qdo current vector
i_r	Current in the dc link, A
I_r	RMS rotor current referred to the stator, A
I_{r1}	Stator-referred fundamental rotor-phase current, A
I_{rms}	RMS armature current, A
I_{rn}	Normalized stator-referred rotor-phase current, p.u.
I_{rr}	Actual RMS rotor-phase current, A
I_{rr1}	Fundamental rotor-phase current, A
I_s	RMS stator-phase current, A
I_s	Steady-state source current to phase-controlled converter, A
I_{sc}	RMS stator-phase current when the rotor is locked, A
I_{sc}	Short-circuit current, A
i_{sn}^*, i_{rn}^*	Normalized stator- and rotor-current phasors in the arbitrary reference frame, p.u.
i_T	Torque-producing component of the stator-current phasor, A
I_T^*	Reference-torque-producing component of stator-current phasor, A

i_T, I_T^*	Steady-state values of i_T , and i_T^* in Chapters 8 and 9
I_T	RMS current of the chopper switch, A, in Ch. 4
I_{Tr}, I_{Trn}	Rated values of I_T in SI and normalized units
I_{ts}	Power-switch current (RMS), A
i_{xyz}^*	Current with superscript s for stator, r for rotor, e for synchronous and c for arbitrary reference frames, and subscript xy for q and d axes and n for normalized unit. Without subscript n , the variable is in SI units.
i_α, i_β	Two-phase instantaneous currents, A
$i_{\alpha m}, i_{\beta m}$	Currents in the new rotor reference frames, A
J	Total moment of inertia, $\text{Kg}\cdot\text{m}^2$
J_l	Moment of inertia of load, $\text{Kg}\cdot\text{m}^2$
J_m	Moment of inertia of motor, $\text{Kg}\cdot\text{m}^2$
K^*	Ratio between control-voltage and rotor-speed references, $\text{V}/\text{rad}/\text{sec}$
K_b	Induced-emf constant, $\text{V}/(\text{rad}/\text{sec})$
K_f	Control-voltage to stator-frequency converter, Hz/V
K_c	Gain of the current controller
K_i	Equivalent gain of the current to its reference
K_{ii}	Integral gain of the current controller
K_{is}	Integral gain of the speed controller
k_m	Ratio between mutual and selfinductances per phase in PMBDC machine
K_{pi}	Proportional gain of the current controller
K_{ps}	Proportional gain of the speed controller
K_r	Converter gain, V/V
k_f	Rotor coupling factor
K_s	Gain of the speed controller
k_s	Stator coupling factor
K_t	Torque constant, $\text{N}\cdot\text{m}/\text{A}$
K_{te}	Torque constant of induction motor, $\text{N}\cdot\text{m}/\text{A}$
K_{tg}	Gain of the tachogenerator, $\text{V}/\text{rad}/\text{s}$
K_{vf}	Ratio between stator-phase voltage and stator frequency, V/Hz
k_w	Winding factor
k_{w1}	Stator winding factor
k_{w2}	Rotor winding factor
L, M	Stator self and mutual inductances in PM brushless machines, H
L_a	DC machine armature inductance, H
L_b	Base inductance, H
L_{eq}	Total leakage inductance, H
L_f	DC machine field inductance, H
L_f	Filter inductance, H
L_i	Interphase inductance, H
L_{lr}	Stator-referred rotor-leakage inductance per phase, H
L_{lrr}	Actual rotor leakage inductance per phase, H
L_{ls}	Stator-leakage inductance per phase, H
L_m	Magnetizing Inductance per phase, H
L_o	Value of inductor in energy-recovery chopper in C-dump topology, H
L_q, L_d	Quadrature- and direct-axis selfinductances, H
L_{qn}, L_{dn}	Normalized quadrature- and direct-axis selfinductances, p.u.
L_{qq}, L_{dd}	Selfinductance of the stator q and d axis windings, H
L_r	Stator-referred rotor selfinductance per phase, H
L_r^*	Vector-controller instrumented L_r
L_s	Stator selfinductance per phase, H
L_s^*	Vector-controller instrumented L_s
L_{sc}	Short circuit inductance, A

L_{sc}	DC series machine field inductance, H
L_{sh}	DC shunt machine field inductance, H
L_{xy}	Mutual inductance between windings given by the subscripts x and y , H
$L_{\alpha\alpha}$	Selfinductance of the rotor α axis windings, H
$L_{\beta\beta}$	Selfinductance of the rotor β axis windings, H
M	Mutual inductance between field and armature windings in dc machine, H
m	Modulation ratio
N_l	Number of turns/phase with half-wave converter control
N_f	Number of turns in the field winding in dc machines
n_r	Rotor speed, rpm
n_s	Stator field speed or synchronous speed, rpm
n_t	Turns ratio of the transformer
o	subscript ending with o in a variable indicates its steady-state operating point value
p	Differential operator
P	Number of Poles
P_l	Air gap power with half wave converter, rad/sec
p_{ln}	Dominant-harmonic resistive losses, W
P_a	Air gap power, W
P_a^*	Air gap power reference, W
P_{an}	Normalized air gap power, p.u.
P_{av}	Average input power, W
P_b	Base power, W
P_c	Armature resistive losses, W
P_{cl}	Machine copper loss with half wave converter operation, W
P_{co}	Core losses, W
P_{c6n}	Normalized sixth harmonic armature copper losses, p.u.
P_{cn}	Normalized armature resistive losses, p.u.
P_{ex}	Energy savings using phase control, W
P_{fw}	Friction and windage losses, W
P_i	Input power, W
P_t	Instantaneous input power, W
P_m	Mechanical power output, W
P_o	Output power, W
P_{on}	Normalized power output, p.u.
P_{rc}	Per-phase rotor resistive losses in ac machines, W
P_s	Shaft power output, W
P_{sc}	Per-phase copper losses in ac machines, W
P_{sl}	Slip power, W
P_{st}	Stray losses in ac machines, W
P_{VA}	Apparent power, VA
Q_i	Reactive power, VAR
R_l	Stator resistance/phase with half-wave converter operation, W
R_a	DC machine armature resistance, Ω
R_{ac}	Changed armature resistance, Ω
R_{an}	Normalized armature resistance, p.u.
R_c	Core-loss resistance, Ω
R_d, R_q	Stator d and q axis winding resistances, Ω
R_{ex}	Braking resistance, Ω
R_f	DC machine field resistance, Ω Ch. 2 and 3
R_f	Dc link filter resistance, Ω
R_{im}	Induction motor equivalent resistance per phase, Ω
R_r	Stator-referred rotor-phase resistance, Ω
R_{rn}	Normalized stator-referred rotor resistance per phase, p.u.
R_s	Stator resistance per phase, Ω

R_{sc}	DC series machine field resistance, Ω
R_{sh}	DC shunt machine field resistance, Ω
R_{sn}	Normalized stator resistance per phase, p.u.
R_α, R_β	Rotor α and β axis winding resistances, Ω
s	Laplace operator and slip in induction machines
S_a, S_b, S_c	a, b, c phase-switching states of the inverter
s_n	n^{th} -harmonic slip
s_r	Rated slip
T	Carrier period time, s
t	Time, s
T_i	Number of turns in phase A
T_1, T_2	Electrical time constants of the motor, s
T_{1e}	Stator effective turns/phase
T_{2e}	Rotor effective turns/phase
T_a	DC machine armature time constant, s
T_{abc}	Transformation from abc to qdo axes
T_{abc}^s	Transformation from abc to qdo variables in the stationary reference frames
T_{av}	Average torque, N.m
T_b	Base torque, N.m
T_c	Time constant of the current controller
T_c	Transformation from stator arbitrary qd variables to stationary qd variables
T_c	Air gap or electromagnetic torque, N.m.
T_c^*	Torque reference, N.m
T_{c1}	First-harmonic air gap torque, N.m
T_{c6}	Sixth-harmonic torque, N.m
T_{c6n}	Normalized sixth-harmonic torque, p.u.
T_{ec}	Torque reference generated by speed error, N.m
T_{ef}	Maximum air-gap torque generated with the voltage and current constraints, N.m
T_{ehn}	Normalized n^{th} harmonic torque, p.u.
T_{en}	Normalized air gap torque, p.u.
T_{en}^*	Normalized torque reference, p.u.
T_{er}	Rated air gap torque, N.m
T_l	Load torque, N.m.
T_{in}	Normalized load torque, p.u.
T_m	Mechanical time constant, s
T_{max}	Maximum torque limit, N.m
T_o	Output torque, N.m.
t_{off}	Chopper off-time, s
t_{on}	Chopper on-time, s
T_r	Converter time delay, s
T_r	Rotor time constant in Ch.8, s
T_r^*	Vector-controller instrumented rotor time constant, s
T_s	Time constant of the speed controller
t_x	Armature current-conduction time, s
$T_{\alpha\beta}$	Transformation from $\alpha\beta$ axes to dq axes
T_w	Time constant of the speed filter, s
u	Input vector
v	Applied voltage to armature of the DC machine, V
v_a^*	Peak value of the carrier signal, V
V_{6p}	Peak sixth-harmonic voltage, V
V_{ab}, V_{bc}, V_{ca}	RMS line-to-line voltages between a and b , b and c and c and a phases, V
v_{ab}, v_{bc}, v_{ca}	Instantaneous line-to-line voltages between a and b , b and c and c and a phases, V
v_{abc}	abc voltage vector

V_{as}	Stator rms voltage input per phase
V_{asn}	Normalized phase a stator voltage, p.u.
V_{ao}, V_{bo}, V_{co}	Inverter midpole voltages, V
V_{ar}	RMS phase a rotor voltage, V
V_{as}	RMS stator phase a voltage, V
V_{av}	Average voltage, V
V_b	Base voltage, V
v_b	Blade velocity, m/s
V_{b3}	Base voltage in 3-phase system, V
v_c	Control voltage, V
V_c	Voltage across the capacitor, V
v_{cf}	Control voltage in the DC field current loop, V
V_{cm}	Maximum control voltage, V
V_{cn}	n^{th} harmonic voltage across the capacitor, V
V_d	Diode peak voltage, V
V_{dc}	DC-link Voltage, V
V_{dc6}	Sixth-harmonic DC-link voltage input, V
V_{dcn}	Normalized DC-link voltage, p.u.
v_{drr}, v_{qrr}	Fictitious rotor voltages in d and q axis, V
v_{ds}, v_{qs}	d and q axis stator voltages, V
V_i	Inverter input voltage, V
V_{i6}	Sixth-harmonic input voltage to the filter, V
V_{i6}	Sixth-harmonic inverter input voltage, V
V_{i6m}	Maximum sixth-harmonic inverter input voltage, V
v_L	Voltage drop across the inductor, V
V_{ll}	Line-to-line AC source voltage, V
V_m	Peak supply voltage, V
v_m	Velocity of the metal flow, m/s
V_n	Normalized applied DC armature voltage, p.u.
V_o	Load voltage, V
V_o	Offset voltage to counter the stator resistive voltage drop, V
V_{o6}	Sixth-harmonic output voltage from the filter, V
V_{on}	Normalized offset voltage, p.u.
v_{os}, v_{or}	Stator and rotor zero-sequence voltages, V
V_{ph}	Fundamental rms phase voltage for the six-stepped voltage, V
v_r	Controlled-rectifier output voltage, V
V_r	Rated armature voltage in DC machines, V
v_R	Voltage drop across the resistor, V
V_{rt}	RMS rotor line voltage, V
V_s	RMS stator-voltage phasor, V
V_{sc}	RMS stator-phase voltage applied when rotor is locked, V
v_{sl}	Slip-speed signal, V
V_{sm}	Maximum slip-speed signal, V
V_{sn}	Normalized stator-voltage phasor, V
v_{tg}	Tachogenerator output voltage, V
V_{ts}	Peak power-switch voltage, V
V_{vs}	Sensor output voltage, V
V_{α}, V_{β}	Rotor voltages in α and β axis windings, V
v_{rn}^c	Normalized rotor-voltage phasor in the arbitrary reference frame, p.u.
v_{sn}^c	Normalized stator-voltage phasor in the arbitrary reference frame, p.u.
v_{xy}^r	Voltage with superscript s for stator, r for rotor, e for synchronous and c for arbitrary reference frames, and subscript xy for q and d axes and n for normalized unit. Without subscript n , the variable is in SI units.

X	State-variable vector
$X(0)$	Initial steady-state vector
X_{an}	Normalized armature reactance, p.u.
X_c	Capacitive reactance, Ω
X_{eq}	Total leakage reactance per phase
X_{im}	Induction-machine equivalent reactance per phase, Ω
X_l	Inductive reactance, Ω
X_{ln}	n^{th} harmonic reactance, Ω
X_{lr}	Stator-referred rotor-leakage reactance per phase, Ω
X_{lrn}	Normalized stator-referred rotor-leakage reactance per phase, p.u.
X_{ls}	Stator-leakage reactance per phase, Ω
X_{lsn}	Normalized stator-leakage reactance per phase, p.u.
X_m	Magnetizing reactance per phase, Ω
X_{mn}	Normalized magnetizing reactance per phase, p.u.
X_{rn}	Normalized stator-referred rotor selfreactance per phase, p.u.
X_{sn}	Normalized stator selfreactance per phase, p.u.
Z	Armature conductors
Z_a	Armature impedance of DC machine, Ω
Z_6	Sixth-harmonic impedance, Ω
Z_{an}	Normalized armature impedance, p.u.
Z_{eq}	Equivalent impedance, Ω
Z_{eqn}	Normalized equivalent impedance
Z_{im}	Induction machine equivalent impedance per phase, Ω
Z_n	n^{th} -harmonic armature impedance, Ω
Z_{sc}	Stator-phase short-circuit impedance, Ω
Δi	Hysteresis-current window, A
ΔI_{dc}	Ripple current in DC link, A
α	Triggering-angle delay, rad
α	Ratio of actual to controller-instrumented rotor time constant in Chapters 8 and 9
α_c	Critical triggering-angle delay, rad
α_f	Triggering angle in the DC field converter, rad
β	Impedance angle of DC machine, rad in Chapter 3; current-conduction angle, rad in Chapter 6
β	Ratio of actual to controller-instrumented mutu inductance in Chapters 8 and 9
δ	Preceding a variable indicates small-signal variation
δ	Torque angle in synchronous machine, rad in Ch. 9
$\delta_\alpha, \delta_\beta$	Error in stator α and β currents in PMSM, A
δR_a	Change in armature resistance, Ω
$\delta \theta$	Error in rotor position, rad
$\delta(\theta)$	Impulse function
δT_c	Air gap torque hysteresis window, N·m
$\delta \lambda_s$	Stator flux-linkages hysteresis window, V·s
$\delta \omega_m$	Change in rotor speed, rad/sec
η	Efficiency
ϕ	Power-factor angle, rad
ϕ_f	Fundamental power-factor angle or displacement angle, rad
ϕ_f	Field flux, Wb
ϕ_{fn}	Normalized field flux, p.u.
ϕ_{fr}	Rated field flux in DC machine, Wb
ϕ_m	Peak Mutual flux, Wb
ϕ_{mr}	Angle between the mutual flux and rotor current, rad
ϕ_o	No-load power-factor angle, rad
γ	Current-conduction angle, rad
λ_1, λ_2	Poles of the DC machine

λ_{af}	Mutual flux linkages due to rotor magnets, V-s
λ_b	Base flux linkages, V-s
λ_{er}	Stator flux error, V-s
λ_m	Mutual air gap flux linkages, V-s
λ_{mn}	Normalized mutual flux linkages, p.u.
$\lambda_{os}, \lambda_{or}$	Stator and rotor zero sequence flux linkages, V-s
λ_p	Peak mutual flux linkages from rotor magnets, V-s
$\lambda_{qr}, \lambda_{dr}$	Rotor flux linkages in q and d axes, V-s
$\lambda_{qs}, \lambda_{ds}$	Stator flux linkages in q and d axes, V-s
λ_r	Rotor flux-linkages phasor, V-s
γ_{rn}^c	Normalized rotor flux linkages in the arbitrary reference frame, p.u.
γ_{sn}^c	Normalized stator flux linkages in the arbitrary reference frame, p.u.
μ	Overlap angle, rad
θ_c	Arbitrary lag angle between q axis and a phase windings, rad
θ_f	Rotor flux linkages or field angle, rad
θ_{fs}	Stator flux-linkages phasor angle, rad
θ_r	Rotor position with respect to d -axis, rads
θ_s	Stator-current phasor angle, rad
θ_{sl}	Slip angle, rad
θ_T	Torque angle, rad
ρ	Saliency ratio, i.e., between q and d axes selfinductances
τ_{hw}	Stator time constant with half-wave converter operation, s
τ	Normalized time, s
τ_c	Time lag between stator current and its command, s
τ_r	Rotor time constant, s
τ_r'	Transient rotor time constant, s
τ_s	Stator time constant, s
τ_s'	Transient stator time constant, s
ω_l	Speed with half-wave converters in Chap. 9, rad/sec
ω_b	Base angular frequency, rad/sec
ω_c	Carrier angular frequency, rad/sec; also, the speed of arbitrary reference frames in induction machines
ω_{cn}	Normalized arbitrary reference frames speed, p.u.
ω_m	Rotor speed, rad/sec
ω_{rn}^*	Normalized speed reference, p.u.
ω_{max}	Normalized maximum rotor speed, p.u.
ω_{mc}	Steady-state speed with increased armature resistance, rad/sec
ω_{mn}	Normalized rotor speed, p.u.
ω_{mo}	Steady-state operating speed, rad/sec
ω_{mr}	Speed signal from the output of speed filter, V
ω_r	Electrical rotor speed, rad/sec
ω_r^*	Speed reference, V
ω_{rm}	Speed of model rotor frames in Chap. 9, rad/sec
ω_{rn}	Normalized rotor speed, p.u.
ω_s	Supply angular velocity, rad/sec
ω_s^*	Stator angular frequency reference, rad/sec
ω_{sl}	Slip speed or slip angular frequency, rad/sec
ω_{sl}^*	Slip-speed reference, rad/sec
ω_r	Rated stator angular frequency, rad/sec
ψ_{os}, ψ_{or}	Modified zero-sequence stator and rotor flux linkages, V-s
ψ_{qr}, ψ_{dr}	Modified rotor flux linkages in q and d axes, V-s
ψ_{qs}, ψ_{ds}	Modified stator flux linkages in q and d axes, V-s
σ	Leakage coefficient of the induction machine
σ	Mutual inductances

ELECTRIC MOTOR DRIVES

Modeling, Analysis, and Control

C H A P T E R 1

Introduction

1.1 INTRODUCTION

The utility power supply is of constant frequency, and it is 50 or 60 Hz. Since the speed of ac machines is proportional to the frequency of input voltages and currents, they have a fixed speed when supplied from power utilities. A number of modern manufacturing processes, such as machine tools, require variable speed. This is true for a large number of applications, some of which are the following:

- (i) Electric propulsion
- (ii) Pumps, fans, and compressors
- (iii) Plant automation
- (iv) Flexible manufacturing systems
- (v) Spindles and servos
- (vi) Aerospace actuators
- (vii) Robotic actuators
- (viii) Cement kilns
- (ix) Steel mills
- (x) Paper and pulp mills
- (xi) Textile mills
- (xii) Automotive applications
- (xiii) Underwater excavators, mining equipment, etc.
- (xiv) Conveyors, elevators, escalators, and lifts
- (xv) Appliances and power tools
- (xvi) Antennas

The introduction of variable-speed drives increases the automation and productivity and, in the process, efficiency. Nearly 65% of the total electric energy produced

in the USA is consumed by electric motors. Decreasing the energy input or increasing the efficiency of the mechanical transmission and processes can reduce the energy consumption. The system efficiency can be increased from 15 to 27% by the introduction of variable-speed drive operation in place of constant-speed operation. It would result in a sizable reduction in the annual energy bill of approximately \$90 billion in USA. That many companies' profits, in recent times, stem mainly from saving in their energy bills is to be noted. The energy-saving aspect of variable-speed drive operation has the benefits of conservation of valuable natural resources, reduction of atmospheric pollution through lower energy production and consumption, and competitiveness due to economy. These benefits are obtained with initial capital investment in variable-speed drives that can be paid off in a short time. The payback period depends on the interest rate at which money is borrowed, annual energy savings, cost of the energy, and depreciation and amortization of the equipment. For a large pump variable-speed drive, it is estimated that the payback period is nearly 3 to 5 years at the present, whereas the total operating life is 20 years. That amounts to 15 to 17 years of profitable operation and energy savings with variable-speed drives.

This section briefly introduces power devices, device switching and losses, motor drive and its subsystems, efficiency and machine rating computation, and power converters, controllers, and load. The scope of the book is included.

1.2 POWER DEVICES AND SWITCHING

1.2.1 Power Devices

Since the advent of semiconductor power switches, the control of voltage, current, power, and frequency has become cost-effective. The precision of control has been enhanced by the use of integrated circuits, microprocessors, and VLSI circuits in control circuits. Some of the popular power-switching devices, their symbols, and their capabilities are described below. The device physics and their functioning in detail are outside the scope of the text, and the interested reader is referred to other sources.

- (i) **Power Diode:** It is a PN device. When its anode potential is higher than the cathode potential by its on-state drop, the device turns on and conducts current. The device on-state voltage drop is typically 0.7 V. When the device is reverse biased, i.e., the anode is less positive than the cathode, the device turns off and goes into blocking mode. The current through the diode goes to zero and then reverses and then resurfaces to zero during the turn-off mode, as shown in Figure 1.1. The reversal of current occurs because the reverse bias leads to the reverse recovery of charge in the device. The minimum time taken for the device to recover its reverse voltage blocking capability is t_{rr} , and the reverse recovery charge contained in the diode is Q_{rr} , shown as the area during the reverse current flow. The diode does not have forward voltage blocking capability beyond its on-state drop. The power diode is available in ratings of

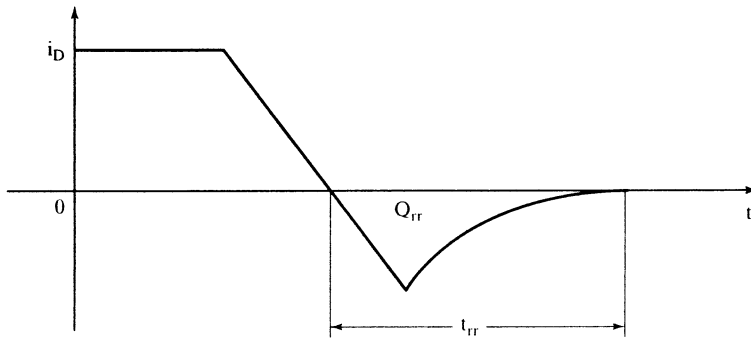


Figure 1.1 Diode current during turn-off

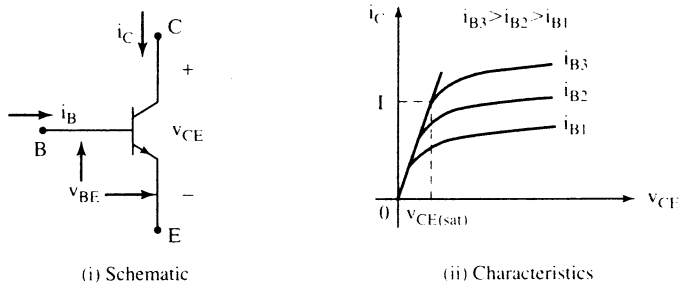


Figure 1.2 Power-transistor schematic and characteristics

kA and kV, and its switching frequency is usually limited to line frequency. Power diodes are used in line rectifier applications.

For fast switching applications, fast-recovery diodes with reverse recovery times in tens of nanoseconds with ratings of several 100 A, at several 100 V, but with a higher on-state drop of 2 to 3 V are available. They are usually used in fast-switching rectifiers with voltages higher than 60 to 100 V and in inverter applications. In case of low-voltage switching applications of less than 60 to 100 V, Schottky diodes are used. They have on-state drop of 0.3 V, thus enabling higher efficiency in power conversion compared to the fast-recovery diodes and power diodes.

- (ii) **Power Transistor:** It is a three-element device, with NPN being the more prevalent. The device can be turned on and off with base current. The device schematic and its characteristics are shown in Figure 1.2. The preferred mode of operation in power circuits is that the transistor be in quasi-saturation, i.e., at knee operating point, during its conduction state. Then it can be pulled back into nonconduction state in shorter time. This device does not have reverse voltage blocking capability. The maximum available rating at present for the device is 1000 A, 1400 V with on-state drop of 2 V. Its switching frequency is very high for the bipolar power transistor, with a lower current gain of 4 to 10, and much in the region of 2 to 6 kHz for Darling power transistors

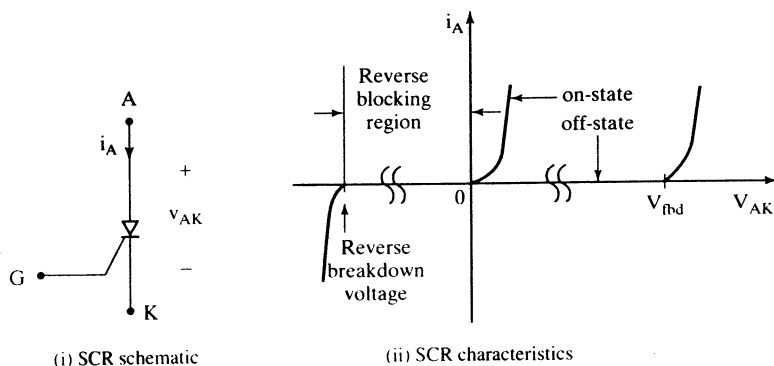


Figure 1.3 SCR schematic and characteristics

with current gain of 100 to 200. These devices are not used very much in newer products.

- (iii) **Silicon Controlled Rectifier (SCR) or Thyristor:** It is a four-element (PNPN) device with three junctions. The power electronics revolution started with the invention of this device in 1956. Its symbol and characteristics are shown in Figure 1.3. The device is turned on with a current signal to the gate, and when the device is forward biased, i.e., the anode is at a higher potential than the cathode by the on-state drop of 1 to 3 V. The device can be turned off only by reverse biasing the device, i.e., reversing the voltage across its anode and cathode. During the reverse biasing, known as commutation, the device behaves like a diode. A negative voltage has to be maintained across the device for a period greater than the reverse recovery time for it to recover its forward blocking voltage capability. The device can also block negative voltages and, beyond a certain value, the device will break down and conduct in the reverse direction. This device is very similar to a power diode but has the capability to hold off its conduction in the forward-biased mode until the gate signal is injected. Its maximum ratings are 6 to 8 kA, 12 kV, with on-state voltage drop ranging from 1 to 3 V. The device is used only in HVDC rectifiers and inverters and in large motor drives with ratings higher than 30 MW. The switching frequency of the device is very limited, to 300 to 400 Hz, and the auxiliary circuit for its turn-off has caused other power devices to displace this device in all applications other than those mentioned. Variations of the device are plentiful; some that belong to this family are the following:

- (a) Inverter-grade SCR
- (b) Light-activated thyristors
- (c) Asymmetrical SCR
- (d) Reverse-conducting thyristor
- (e) MOS-controlled thyristor
- (f) Gate turn-off thyristor (GTO)

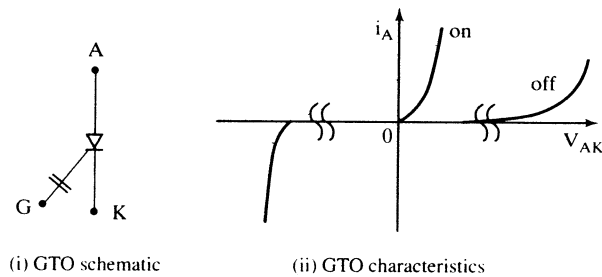


Figure 1.4 The schematic and characteristics of the GTO

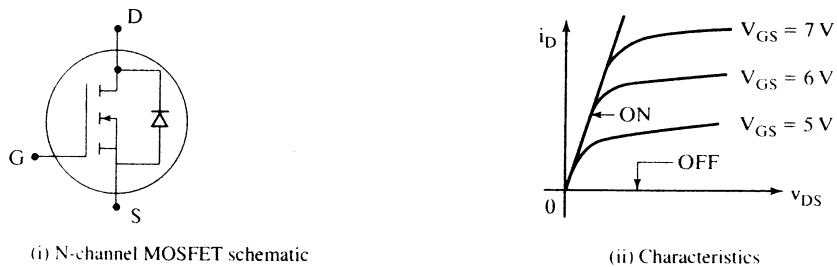


Figure 1.5 Schematic and characteristics of the N-channel MOSFET

- (iv) **Gate Turn-Off Thyristor (GTO):** It is a thyristor device with gate turn-on and gate turn-off capability. Its symbol and characteristics are shown in Figure 1.4. The device comes with maximum ratings of 6 kA, 6 kV, with an on-state voltage drop of 2 to 3 V. The maximum switching frequency is 1 kHz, and the device is used mainly in high-power inverters.
- (v) **MOSFET:** This device is a class of field-effect transistor requiring lower gate voltages for turn-on and turn-off and capable of higher switching frequency in the range of 30 kHz to 1 MHz. The device is available at 100 A at 100 to 200 V and at 10 A at 1000 V. The device behaves like a resistor when in conduction and therefore can be used as a current sensor, thus eliminating one sensor device in a drive system. The device always comes with an anti-parallel body diode, sometimes referred to as a *parasitic* diode, that is not ultra-fast and has a higher voltage drop. The device symbol for an N-channel MOSFET and its characteristics are shown in Figure 1.5. The device has no reverse voltage blocking capability.
- (vi) **Insulated Gate Bipolar Transistor:** It is a three-element device with the desirable characteristics of a MOSFET from the viewpoint of gating, transistor in conduction, and SCR/GTO in reverse voltage blocking capability. Its symbol is given in Figure 1.6. The currently available ratings are 1.2 kA at 3.3 kV and 0.6 kA at 6.6 kV, with on-state voltage drop of 5 V. Higher currents at reduced voltages with much lower on-state voltage drops are available. It is expected that further augmentation in the maximum current and voltage ratings will

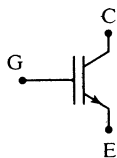


Figure 1.6 IGBT symbol

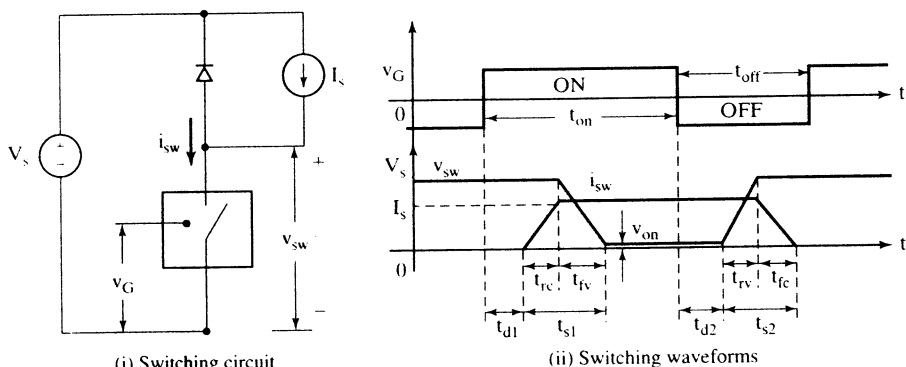


Figure 1.7 Switching circuit and waveforms

occur in the future. The switching frequency is usually around 20 kHz for many of the devices, and its utilization at high power is at low frequency, because of switching loss and electromagnetic-interference concerns.

1.2.2 Switching of Power Devices

The understanding of device switching in transient is of importance in the design of the converter as it relates to its losses, efficiency of the converter and the motor drive system, and thermal management of the power converter package. The transient switching of the devices during turn-on and turn-off is illustrated in this section by considering a generic device. Ideal current and voltage sources and power devices are assumed for this illustration. The circuit for illustration is shown Figure 1.7. The switching device is gated on and the current in the device increases from zero to I_s after a turn-on delay time t_{d1} . The current transfers from the diode linearly in time t_{rc} , which is the rise time of the current. During this rise time, the diode is conducting, and therefore the voltage across the switching device is the source voltage, V_s . When the current is completely transferred from the diode to the switching device, the voltage drop across the diode rises from zero to the source voltage and the voltage falls at the same time across the switching device in t_{fv} . The sum of the current rise and voltage fall times is the turn-on switching transient time; note that, during this time, the device losses are very high. During the conduction, the voltage across the device is its on-state voltage drop and the power loss is smaller.

When the gating signal goes to the turn-off condition, the switching device responds with a turn-off delay time of t_{d2} . Then, the device voltage rises to V_s in t_{rv} , which forward biases the diode and initiates the current transfer from the switching device to the diode. The current transfer is completed in t_{fc} . The sum of the voltage rise time across the switching device and current fall time is the turn-off transient time, during which the device loss is very high.

From this illustration, the losses in the switching device are approximately derived as follows:

$$(a) \text{ Conduction energy loss, } E_{sc} = I_s V_{on} [t_{on} + t_{d2} - t_{s1} - t_{d1}] \quad (1.1)$$

$$(b) \text{ Sum of turn-on and turn-off energy loss, } E_{st} \cong 0.5 V_s I_s [t_{s1} + t_{s2}] \quad (1.2)$$

$$(c) \text{ Total power loss, } P_{sw} = \frac{E_{st} + E_{sc}}{t_{on} + t_{off}} = f_c (E_{sc} + E_{st}) \quad (1.3)$$

$$(d) \text{ Switching frequency, } f_c = \frac{1}{t_{on} + t_{off}} \quad (1.4)$$

Note that the power loss is averaged over a period of the switching period, and total power loss is the sum of the conduction and switching losses in the switching device. Similarly, the power losses in the diode can be derived. The switching losses are proportional to the switching frequency and to the product of the source voltage and load current. Note that, in general, the switching times are much smaller than the conduction time, and therefore the switching losses are less than (or at most equal to) the conduction loss in the switching device. Low switching frequency is preferred because of lower switching losses but contributes to poor power quality. Higher switching frequency enables voltage and current waveform shaping and reduces the distortion but invariably is followed by higher switching losses.

The switching illustrated is known as hard switching; current and voltage transitions occur in the device at full source voltage and current, respectively, during turn-on and turn-off periods. Resonant and soft-switching circuits enable switching device transitions at zero voltage and current, reducing or almost eliminating the switching losses, but these circuits are not economical at present and so are not considered in motor-drive applications or in this text any further.

The power switches are used in the circuits to control energy flow from source to load and vice versa. This is known as static power conversion. The details of power conversion are not the objectives of this text, and many good texts are available on this topic. Some references are listed at the end of this chapter. As and when necessary, a brief description of a power converter is included in this text.

1.3 MOTOR DRIVE

A modern variable-speed system has four components:

- (i) Electric machines—ac or dc
- (ii) Power converter—Rectifiers, choppers, inverters, and cycloconverters

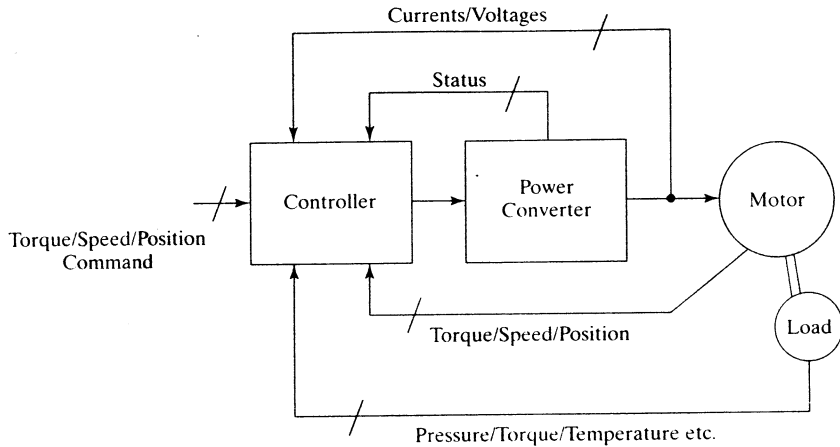


Figure 1.8 Motor-drive schematic

- (iii) Controllers—Matching the motor and power converter to meet the load requirements
- (iv) Load

This is represented in the block diagram shown in Figure 1.8. A brief description of the system components is given in the following sections.

1.3.1 Electric Machines

The electric machines currently used for speed control applications are the following:

- (i) dc machines—Shunt, series, compound, separately-excited dc motors, and switched reluctance machines;
- (ii) ac machines—Induction, wound-rotor synchronous, permanent-magnet synchronous, synchronous reluctance, and switched reluctance machines.
- (iii) Special machines—Switched reluctance machines.

All the machines are commercially available from fractional-kW to MW ranges except permanent-magnet synchronous, synchronous reluctance, and switched reluctance machines, which are available up to 150 kW level. The latter machines are available at higher power levels but would be expensive from a commercial point of view, because they require custom designs. A number of factors go into the selection of a machine for a particular application:

- (i) Cost
- (ii) Thermal capacity
- (iii) Efficiency
- (iv) Torque–speed profile

- (v) Acceleration
- (vi) Power density, volume of the motor
- (vii) Ripple, cogging torques
- (viii) Availability of spare and second sources
- (ix) Robustness
- (x) Suitability for hazardous environment
- (xi) Peak torque capability

They are not uniformly relevant to any one application. Some could take precedence over others. For example, in a position-servo application, the peak torque and thermal capabilities together with ripple and cogging torques are preponderant characteristics for application consideration. High peak torques decrease the acceleration/deceleration times, small cogging and ripple torques help to attain high positioning repeatability, and high thermal capability leads to a longer motor life and a higher loading. In this text, only dc, induction, and permanent-magnet synchronous and brushless dc machines are considered, and their steady-state and dynamic models are derived. Wound-rotor synchronous machines and drives technology is well established, and very little change has occurred in them over the last 40 years. The synchronous reluctance and switched reluctance machines and their drive systems are fairly recent but yet to establish themselves in the market. Very few engineers are involved with these topics, whereas a large number of engineers are employed in the dc, induction, permanent-magnet synchronous, and brushless dc machines industrial sector, and, therefore, only these topics are covered in this text.

Efficiency computation. The interest in energy savings is one of the major motivational factors in the introduction of variable-speed drives in some industries. Therefore, it is prevalent to encounter the efficiency computation for electric motors whenever variable-speed operation is considered. A brief introduction of efficiency computation for electric motors is given in the following.

The power, voltage and speed given in the nameplate details of the motor are its rated values, i.e., for continuous steady-state operation. Notice how the power corresponds to shaft output power. The output or shaft torque, T_o , of the machine is calculated as

$$\left. \begin{aligned} T_o &= \frac{\text{Power, W}}{\text{Speed, rad/sec}} = \frac{P_o \times 745.6}{\omega_m}, \text{ N}\cdot\text{m} \\ \omega_m &= \frac{2\pi N_r}{60}, \text{ rad/sec} \end{aligned} \right\} \quad (1.5)$$

where P_o is the output power in hp, N_r is speed in rpm, and ω_m is the speed in rad/sec.

The internal torque, T_e , of the motor is the sum of the output torque and the shaft torque losses, viz., friction and windage torques. Friction is contributed by the bearings (usually proportional to speed) and windage by the effects of the circulating air on the rotating parts (proportional to square of the speed). Let the shaft

power losses be denoted by P_{sh} and loss torque by T_{lo} . The internal torque then is given by

$$T_e = T_o + T_{lo}, \text{ N}\cdot\text{m} \quad (1.6)$$

This internal torque is known as air gap or electromagnetic torque, because it is the torque developed by the motor in the air gap through electromagnetic coupling. The air gap torque is developed from the power crossing over to the air gap from the armature (usually the rotor in dc machines, the stator in ac machines) and is known as air gap power, P_a . In general, it is obtained from the armature input, P_i , and its loss, P_l , due to armature resistance (known as copper losses), core losses, and unidentifiable losses (known as stray losses, ranging from 0.5 to 1% of output power).

Note that air gap torque from air gap power is calculated as

$$T_e = \frac{P_a}{\omega_m}, \text{ N}\cdot\text{m} \quad (1.7)$$

The efficiency, then, is computed as

$$\text{Efficiency, } \eta = \frac{\text{Output power}}{\text{Input power}} = \frac{P_o}{P_i} \quad (1.8)$$

For particular motors, the details of the computation are shown in respective chapters. Note that the output torque is equal to the load torque, T_l . If shaft loss torque is neglected, note that the electromagnetic torque is equal to load torque.

Motor rating. A typical electric train's torque and speed profiles for one cycle are shown in Figure 1.9. It is seen that neither the load torque nor the speed is

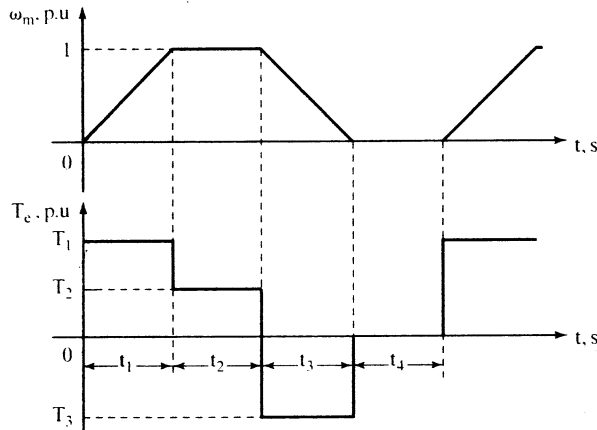


Figure 1.9 An electric train's torque and speed profiles for one cycle

constant; both vary. Such loads are encountered in practice. The selection of motor torque and power ratings, then, is not straightforward, as in the case of constant-speed drives. In the varying-load case, the motor torque and power are selected on the basis of effective torque and power. The effective torque is calculated from the load profile, such as the one shown in Figure 1.9. For example, the effective electromagnetic torque for the load cycle shown in Figure 1.9, discounting the rest period, is obtained as

$$T_e = \sqrt{\frac{T_1^2 t_1 + T_2^2 t_2 + T_3^2 t_3}{t_1 + t_2 + t_3}} \quad (1.9)$$

Similarly, the effective power is calculated from its time history. These calculations yield effective torque and power. Ensuring the effective torque and output power to be well below or equal to the rated values of the machine does not guarantee that the machine is thermally within its designed limits. The thermal rating influences the motor operation. The thermal rating is dependent on the motor losses, and they, in turn, are preponderantly influenced by the effective current in the case of a dc machine but also by additional factors, such as stator frequency and applied voltages, in ac machines. In order to determine whether the machine is operating within the thermal limits of rated power losses, it is necessary to calculate the effective losses in the machine for a load cycle. If the effective losses are lower than the rated losses, then the machine is thermally sound for operation.

Example 1.1

Two electrical machines are considered to drive a load. The torque and speed ratings of the machines are same. The thermal limits for the machines correspond to their rated armature resistive losses. The machines 1 and 2 have the following torque-to-armature current relationships:

$$(1) T_e \propto i^2$$

$$(2) T_e \propto i$$

The load is twice the rated torque for half the time and zero for the other half of the time in a load cycle. Find which machine is suitable for the application.

Solution By ignoring the shaft-loss torque, the load torque is equal to the electromagnetic torque. Let T_b be the base (rated) torque, I_b the base current, and $I_b^2 R$ the base losses. The effective electromagnetic torque is

$$T_e = 2T_b \sqrt{d} = 2T_b \sqrt{\frac{1}{2}} = \sqrt{2}T_b$$

where d is the duty cycle.

Machine 1: From the relationship that the electromagnetic torque is proportional to square of the current, the current for the load is found as follows:

$$T_b \propto I_b^2$$

$$2T_b \propto I_1^2$$

The armature current for the load is $I_1 = I_b \sqrt{2}$

RMS armature current is $I_{1r} = I_b \sqrt{2} \sqrt{d} = I_b \sqrt{2} \sqrt{\frac{1}{2}} = I_b$

Losses $= I_{1r}^2 R = I_b^2 R = \text{Rated losses}$

Machine 2: As electromagnetic torque is proportional to current, the current for the load is found as follows for this machine:

$$T_b \propto I_b$$

$$2T_b \propto I_2$$

The armature current for the given load is $I_2 = 2I_b$

RMS value of the armature current is $I_{2r} = 2I_b \sqrt{d} = 2I_b \sqrt{\frac{1}{2}} = I_b \sqrt{2}$

Loss $= I_{2r}^2 R = 2I_b^2 R = 2 * \text{Rated losses}$

Therefore, machine 1 alone satisfies the thermal limit of rated losses and hence is suitable for the given application. Note that the effective torque is more than the rated torque, but, still, thermally safe operation is possible for the considered case.

1.3.2 Power Converters

The power converters driving the electrical machines are

- (i) **Controlled rectifiers:** They are fed from single- and three-phase ac mains supply and provide a dc output for control of the dc machines or sometimes input dc supply to the inverters in the case of ac machines.
- (ii) **Inverters:** They provide variable alternating voltages and currents at desired frequency and phase for the control of ac machines. The dc supply input to the inverters is derived either from a battery in the case of the electric vehicle or from a rectified ac source with controlled or uncontrolled (diode) rectifiers. Because of the dc intermediary, known as dc link, between the supply ac source and the output of the inverter, there is no limitation to the attainable output frequency other than that of the power device switching constraints in the inverters.
- (iii) **Cycloconverters:** They provide a direct conversion of fixed-frequency alternating voltage/current to a variable voltage/current variable frequency for the control of ac machines. The output frequency is usually limited from 33 to 50% of the input supply frequency, to avoid distortion of the waveform, and therefore they are used only in low-speed but high-power ac motor drives.

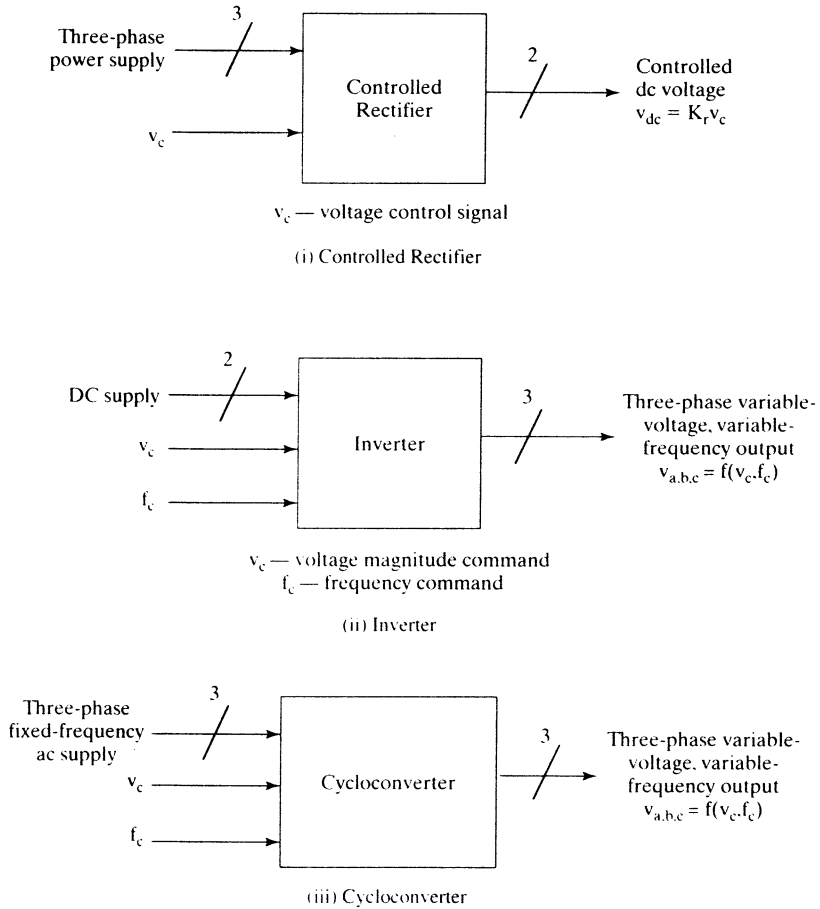


Figure 1.10 Schematics of power converters

These power converters can be treated as black boxes with certain transfer functions. In that case, the referred converters are symbolically represented as shown in Figure 1.10. The transfer functions are derived for various converters in this text.

1.3.3 Controllers

The controllers embody the control laws governing the load and motor characteristics and their interaction. To match the load and motor through the power converter, the controller controls the input to the power converter. Very many control strategies have been formulated for various motor drives and the controllers implement their algorithms. For instance, the control of flux and torque requires a coordination of the field and armature currents in a separately excited dc motor. In the case of an ac induction motor, the same is implemented by coordinating the three stator currents, whereas the

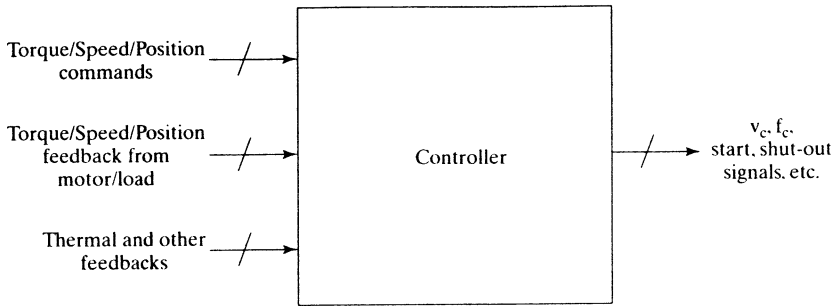


Figure 1.11 Controller schematic

synchronous motor control requires the control of the three stator currents and its field current too. The laws governing their control are complex and form the core of this text.

The schematic of the controller is shown in Figure 1.11. Its input consists of the following:

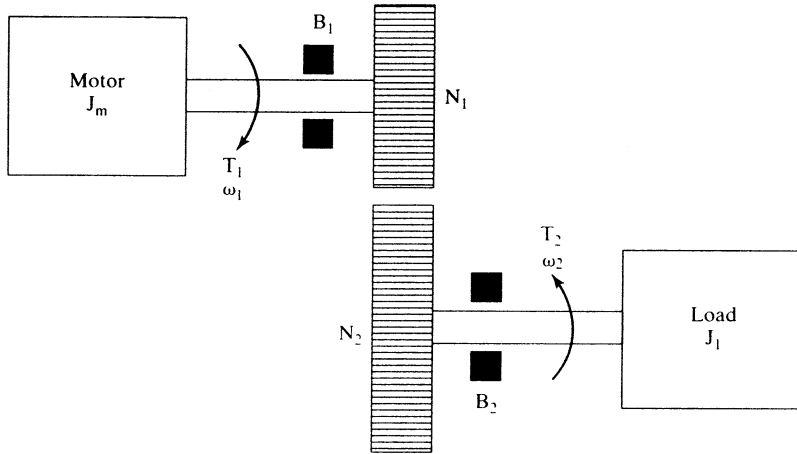
- (i) Torque, flux, speed, and/or position commands
- (ii) Their rate of variations, to facilitate soft start and preserve the mechanical integrity of the load
- (iii) The measured torque, flux, speed, and/or position for feedback control
- (iv) Limiting values of currents, torque, acceleration, and so on
- (v) Temperature feedback and instantaneous currents and/or voltages in the motor and/or converter
- (vi) The constants in the speed and position controllers, such as proportional, integral, and differential gains.

The controller output determines the control signal for voltage magnitude, v_c in the case of inverters, and the control signal for determining the frequency, f_c . These functions can be merged, and only the final gating signals might be directly issued to the bases/gates of the power converter. It may also perform the protection and other monitoring functions and deal with emergencies such as sudden field loss or power failure.

The controllers are realized with analog and integrated circuits. The present trend is to use microprocessors, single-chip microcontrollers, digital signal processors [DSPs], VLSI, and special custom chips also known as application specific ICs [ASICs] to embody a set of functions in the controller. The real-time computational capability of these controllers allows complex control algorithms to be implemented. Also, they lend themselves to software and remote control, hence paving the way to flexible manufacturing systems and a high degree of automation.

1.3.4 Load

The motor drives a load that has a certain characteristic torque-vs.-speed requirement. In general, the load torque is a function of speed and can be written as $T_l \propto \omega_m^x$, where x can be an integer or a fraction. For example, the load torque is



N_1, N_2 — Teeth numbers in the gear
 B_1, B_2 — Bearings and their friction coefficients
 J_l, J_m — Moment of inertia of the motor and load

Figure 1.12 Schematic of the motor-load connection through a gear

proportional to speed in frictional systems such as a feed drive. In fans and pumps, the load torque is proportional to the square of the speed.

In some instances, the motor is connected to the load through a set of gears. The gears have a teeth ratio and can be treated as torque transformers. The motor-gear-load connection is schematically shown in Figure 1.12. The gears are primarily used to amplify the torque on the load side that is at a lower speed compared to the motor speed. The motor is designed to run at high speeds because it has been found that the higher the speed, the lower is the volume and size of the motor. But most of the useful motion is at low speeds, hence the need for a gear in the motor-load connection. The gears can be modeled from the following facts [5]:

- (i) The power handled by the gear is the same on both the sides.
- (ii) Speed on each side is inversely proportional to its tooth number.

Hence,

$$T_1 \omega_1 = T_2 \omega_2 \quad (1.10)$$

$$T_2 = T_1 \left(\frac{\omega_1}{\omega_2} \right) \quad (1.11)$$

and

$$\frac{\omega_1}{\omega_2} = \frac{N_2}{N_1} \quad (1.12)$$

Substituting equation (1.12) into (1.11), we get

$$T_2 = \left(\frac{N_2}{N_1} \right) T_1 \quad (1.13)$$

Note that this is obtained with the assumption of zero losses in the gear, and for greater accuracy the losses have to be modeled. In that case, the difference between T_1 and the loss torque will have to be multiplied by the turns ratio, N_2/N_1 , to obtain the output gear torque T_2 . Similarly to the case of a transformer, the constants of the load as reflected to the motor are written as

$$J_1 (\text{reflected}) = \left(\frac{N_1}{N_2} \right)^2 J_1 \quad (1.14)$$

$$B_2 (\text{reflected}) = \left(\frac{N_1}{N_2} \right)^2 B_2 \quad (1.15)$$

The ratio between N_1 and N_2 is very small, so the reflected inertia and friction constant are negligible in many systems.

Hence, the resultant mechanical constants are

$$J = J_m + \left(\frac{N_1}{N_2} \right)^2 J_1 \quad (1.16)$$

$$B = B_1 + \left(\frac{N_1}{N_2} \right)^2 B_2 \quad (1.17)$$

The torque equation of the motor-load combination is described by

$$J \frac{d\omega_1}{dt} + B\omega_1 = T_1 - T_2(\text{reflected}) = T_1 - \left(\frac{N_1}{N_2} \right) T_2 \quad (1.18)$$

To model the motor drive, it is essential to have a physical model of its load, with the characteristics of friction, inertia, torque-speed profile, and gears and backlash. Modeling of the physical systems can be found in references [8, 11].

1.4 SCOPE OF THE BOOK

This book addresses the study of steady-state and dynamic control of ac and dc machines supplied from power converters and their integration to the load. In due course, the design of controllers and their implementation are considered. The system analysis and design of the motor drive are kept in perspective with regard to current practice. Control algorithms and analysis have been developed to facilitate dynamic simulation with personal computers. Applications of motor drives are illustrated with selections from industrial environment.

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C H A P T E R 2

Modeling of DC Machines

Direct current (dc) machines have been in service for more than a century. Their fortune has changed a great deal since the introduction of the induction motor, sometimes called the ac shunt motor. DC motors have staged a comeback with the advent of the silicon-controlled rectifier used for power conversion, facilitating a wide-range speed control of these motors.

This chapter contains a brief description of the theory of operation of separately-excited and permanent-magnet dc brush motors and of their modeling and transfer functions, an evaluation of steady state and transient responses, and a flow chart for their computation.

2.1 THEORY OF OPERATION

It is known that maximum torque is produced when two fluxes are in quadrature. The fluxes are created with two current-carrying conductors. The flux path is of low reluctance with steel. Figure 2.1 shows such a schematic representation. Coil 1 (called the field winding) wound on the pole produces a flux ϕ_f with an input current i_f . Coil 2 (called the armature winding) on a rotating surface produces a flux ϕ_a with a current i_a . For the given position, the two fluxes are mutually perpendicular and hence will exert a maximum torque on the rotor, moving it in clockwise direction. If it is assumed that the rotor has moved by 180° , coils B_1 , B_2 , and B_3 will be under the south pole and carrying negative current. The torque will be such as to move the rotor in the counterclockwise direction, keeping the rotor in oscillation. In order to have a uniform torque (i.e. unidirectional torque) and a clockwise (or counterclockwise) direction of rotation, the armature winding needs to carry a current of the same polarity underneath a field pole. That is arranged by segmenting the armature coils and connecting them to separate copper bars, called a commutator, mounted on the same shaft as the armature and fed from brushes. The brushes are stationary and hence supplied with currents of fixed polarity. Thus, the commutator segments under a brush will continue to receive a current of fixed polarity.

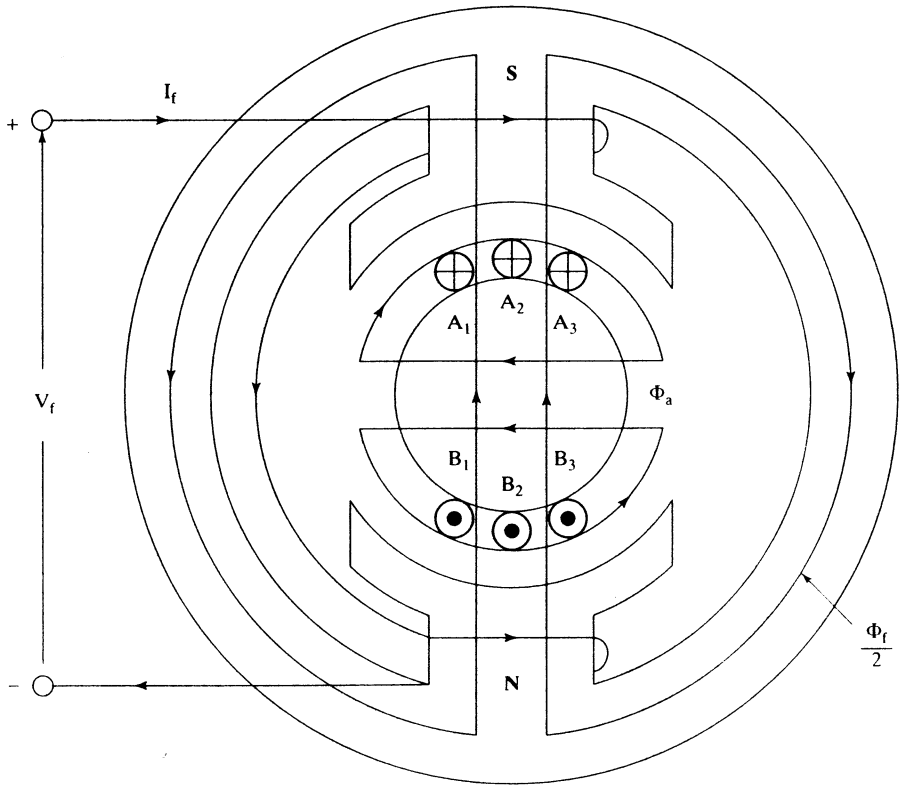


Figure 2.1 Schematic representation of a dc machine

2.2 INDUCED EMF

The expression for induced emf and torque is derived for a machine with P poles, Z armature conductors in a field with a flux per pole of Φ_f and rotating at n_r rpm. From Faraday's law, the induced emf (neglecting the sign) is

$$e = Z \frac{d\Phi_f}{dt} = Z \frac{\Phi_f}{t} \quad (2.1)$$

where t is the time taken by the conductors to cut Φ_f flux lines. Therefore

$$t = \frac{1}{2 \times \text{frequency}} = \frac{1}{2 \left(\frac{P}{2} \right) \left(\frac{n_r}{60} \right)} \quad (2.2)$$

The flux change occurs for each pole pair.

By substituting equation (2.2) in equation (2.1),

$$e = \frac{Z \Phi_f P n_r}{60} \quad (2.3)$$

If the armature conductors are divided into 'a' parallel paths, then

$$e = \frac{Z\phi_r P n_r}{60a} \quad (2.4)$$

There are two possible arrangements of conductors in the armature, wave windings and lap windings. The values of a for these two types of windings are

$$a = \begin{cases} 2 & \text{for wave winding} \\ P & \text{for lap winding} \end{cases} \quad (2.5)$$

It is usual to write the expression (2.4) in a compact form as

$$e = K\phi_r\omega_m \quad (2.6)$$

where $\omega_m = 2\pi n_r/60$ rad/sec and $K = (P/a)Z(1/2\pi)$.

If the field flux is constant, then the induced emf is proportional to the rotor speed and the constant of proportionality is known as the *induced emf* or *back emf* constant. Then the induced emf is represented as

$$e = K_b\omega_m \quad (2.7)$$

where K_b is the induced emf constant, given by

$$K_b = K\phi_r \text{ volt/(rad/sec)} \quad (2.8)$$

and K is a proportionality constant which will be obvious from the following. The field flux is written as the ratio between the field mmf and mutual reluctance,

$$\phi_r = \frac{N_f i_f}{\mathcal{R}_m} \quad (2.9)$$

where N_f is the number of turns in the field winding, i_f is the field current, and $\mathcal{R}_m$ is the reluctance of the mutual flux path. The mutual flux is the resultant of the armature and field fluxes.

By substituting (2.9) into (2.7), the emf constant is obtained as

$$K_b = \frac{KN_f i_f}{\mathcal{R}_m} = M i_f \quad (2.10)$$

where M is the fictitious mutual inductance between armature and field windings given by

$$M = \frac{KN_f}{\mathcal{R}_m} = \frac{P}{\pi} \frac{Z}{2a} \frac{N_f}{\mathcal{R}_m} \quad (2.11)$$

Note that $Z/(2a)$ is the number of turns in the armature per parallel path and together in product with the field winding turns N_f gives the familiar mutual inductance definition. The factor P/π makes it a fictitious inductance. By substituting equation (2.10) in equation (2.7), the induced emf is obtained as

$$e = M i_f \omega_m \quad (2.12)$$

The mutual inductance is a function of the field current and must be taken note of to account for saturation of the magnetic material (stator laminations). For operation within the linear range, mutual inductance is assumed to be a constant in the machine.

2.3 EQUIVALENT CIRCUIT AND ELECTROMAGNETIC TORQUE

The equivalent circuit of a dc motor armature is based on the fact that the armature winding has a resistance R_a , a self-inductance L_a , and an induced emf. This is shown in Figure 2.2. In the case of a motor, the input is electrical energy and the output is the mechanical energy, with an air gap torque of T_e at a rotational speed of ω_m . The terminal relationship is written as

$$v = e + R_a i_a + L_a \frac{di_a}{dt} \quad (2.13)$$

In steady state, the armature current is constant and hence the rate of change of the armature current is zero. Hence the armature voltage equation reduces to

$$v = e + R_a i_a \quad (2.14)$$

The power balance is obtained by multiplying equation (2.14) by i_a :

$$v i_a = e i_a + R_a i_a^2 \quad (2.15)$$

The term $R_a i_a^2$ denotes the armature copper losses and $v i_a$ is the total input power. Hence $e i_a$ denotes the effective power that has been transformed from electrical to

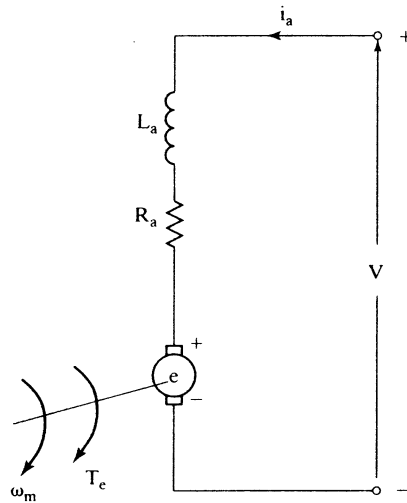


Figure 2.2 Equivalent circuit of a dc motor armature

mechanical form, hereafter called the air gap power, P_a . The air gap power is expressed in terms of the electromagnetic torque and speed as

$$P_a = \omega_m T_e = e i_a \quad (2.16)$$

Hence, the electromagnetic torque or air gap torque is represented as

$$T_e = \frac{e i_a}{\omega_m} \quad (2.17)$$

By substituting for the induced emf from (2.7) into (2.17), it is further simply represented as

$$T_e = K_b i_a \quad (2.18)$$

Note that the torque constant is equal to the emf constant if it is expressed in volt-sec/rad for a constant-flux machine. The dc motor drives a load, the modeling of load is considered next, to complete the analysis for the motor-load system.

2.4 ELECTROMECHANICAL MODELING

For simplicity, the load is modeled as a moment of inertia, J , in $\text{kg-m}^2/\text{sec}^2$, with a viscous friction coefficient B_1 in $\text{N-m}/(\text{rad}/\text{sec})$. Then the acceleration torque, T_a , in N-m drives the load and is given by

$$J \frac{d\omega_m}{dt} + B_1 \omega_m = T_e - T_l = T_a \quad (2.19)$$

where T_l is the load torque. Equations (2.13) and (2.19) constitute the dynamic model of the dc motor with load.

2.5 STATE-SPACE MODELING

The dynamic equations are cast in state-space form and are given by

$$\begin{bmatrix} p i_a \\ p \omega_m \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_b}{L_a} \\ \frac{K_b}{J} & -\frac{B_1}{J} \end{bmatrix} \begin{bmatrix} i_a \\ \omega_m \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix} \begin{bmatrix} V \\ T_l \end{bmatrix} \quad (2.20)$$

where p is the differential operator with respect to time. Equation (2.19) is expressed compactly in the form given by

$$\dot{X} = AX + BU \quad (2.21)$$

where $X = [i_a \ \omega_m]^T$, $U = [V \ T_l]^T$, X is the state variable vector, and U is the input vector.

Even though the load torque is a disturbance, for sake of a compact representation it is included in the input vector in this text.

$$A = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_b}{L_a} \\ \frac{K_b}{J} & -\frac{B_l}{J} \end{bmatrix}, B = \begin{bmatrix} \frac{1}{L_a} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix}$$

The roots of the system are evaluated from the A matrix; they are

$$\lambda_1, \lambda_2 = \frac{-\left(\frac{R_a}{L_a} + \frac{B_l}{J}\right) \pm \sqrt{\left(\frac{R_a}{L_a} + \frac{B_l}{J}\right)^2 - 4\left(\frac{R_a B_l}{J L_a} + \frac{K_b^2}{J L_a}\right)}}{2} \quad (2.22)$$

It is interesting to observe that these eigenvalues will always have a negative real part, indicating that the motor is stable on open-loop operation.

2.6 BLOCK DIAGRAMS AND TRANSFER FUNCTIONS

Taking Laplace transforms of equations (2.13), (2.19) and neglecting initial conditions, we get

$$I_a(s) = \frac{V(s) - K_b \omega_m(s)}{R_a + sL_a} \quad (2.23)$$

$$\omega_m(s) = \frac{K_b I_a(s) - T_l(s)}{(B_l + sJ)} \quad (2.24)$$

In block-diagram form, the relationships are represented in Figure 2.3. The transfer functions $\frac{\omega_m(s)}{V(s)}$ and $\frac{\omega_m(s)}{T_l(s)}$ are derived from the block diagram. They are

$$G_{\omega V}(s) = \frac{\omega_m(s)}{V(s)} = \frac{K_b}{s^2(JL_a) + s(B_l L_a + J R_a) + (B_l R_a + K_b^2)} \quad (2.25)$$

$$G_{\omega l}(s) = \frac{\omega_m(s)}{T_l(s)} = \frac{-(R_a + sL_a)}{s^2(JL_a) + s(B_l L_a + J R_a) + (B_l R_a + K_b^2)} \quad (2.26)$$

The separately-excited dc motor is a linear system, and hence the speed response due to the simultaneous voltage input and load torque disturbance can be written as a sum of their individual responses:

$$\omega_m(s) = G_{\omega V}(s)V(s) + G_{\omega l}(s)T_l(s) \quad (2.27)$$

Laplace inverse of $\omega_m(s)$ in equation (2.27) gives the time response of the speed for a simultaneous change in the input voltage and load torque. The treatment so far is

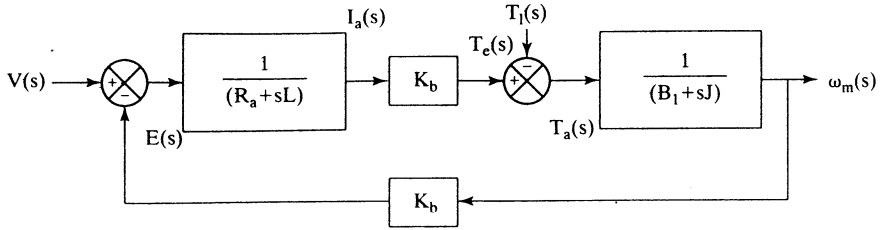


Figure 2.3 Block diagram of the dc motor

based on a dc motor obtaining its excitation separately. There are various forms of field excitation, discussed in the following section.

Example 2.1

A dc motor whose parameters are given in example 2.3 is started directly from a 220-V dc supply with no load. Find its starting speed response and the time taken to reach 100 rad/sec.

Solution

$$\frac{\omega(s)}{V(s)} = G_{\omega v}(s) = \frac{K_b}{s^2(JL_a) + s(B_1L_a + JR_a) + (B_1R_a + K_b^2)} = \frac{15.968}{s^2 + 167s + 12874}$$

$$V(s) = \frac{220}{s}$$

$$\omega(s) = \frac{3.512 \times 10^6}{s(s^2 + 167s + 12874)}$$

$$\omega(t) = 272.8(1 - 1.47e^{-83.5t} \sin(76.02t + 0.744))$$

The time to reach 100 rad/sec is evaluated by equating the left-hand side of the above equation to 100 and solving for t , giving an approximate value of 10 rad/ms.

2.7 FIELD EXCITATION

The excitation to the field is dependent on the connections of the field winding relative to the armature winding. A number of choices open up, and they are treated briefly in the following sections.

2.7.1 Separately Excited dc Machine

If the field winding is physically and electrically separate from the armature winding, then the machine is known as a separately-excited dc machine, whose equivalent circuit is shown in Figure 2.4. The independent control of field current and armature current endows simple but high performance control on this machine, because the torque and flux can be independently and precisely controlled. The

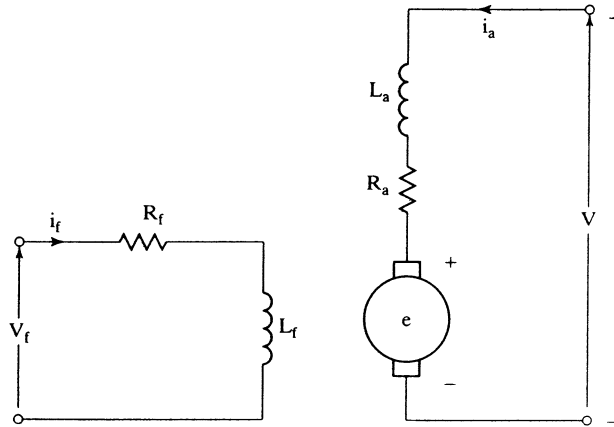


Figure 2.4 Separately-excited dc machine

field flux is controlled only by the control of the field current. Assume that the field is constant. Then the torque is proportional only to the armature current, and, hence, by controlling only this variable, the dynamics of the motor drive system can be controlled. With the independence of the torque and flux-production channels in this machine, it must be noted that it is easy to generate varying torques for a given speed and hence make torque generation independent of the operating speed. This is an important operating feature in a machine: the speed regulation can be zero. The fact that such a feature comes only with feedback control and not simply on open-loop operation is to be recognized.

Example 2.2

A separately-excited dc motor is delivering rated torque at rated speed. Find the efficiency of the motor at this operating point. The details of the machine are as follows: 1500 kW, 600V, rated current = 2650 A, 600 rpm, Brush voltage drop = 2 V, Field power input = 50 kW, $R_a = 0.003645 \Omega$, $L_a = 0.1$ mH, Machine frictional torque coefficient = $15 \text{ N}\cdot\text{m}/(\text{rad}/\text{sec})$. Field current is constant and the armature voltage is variable.

Solution To find the input power, the applied voltage to the armature to support a rated torque and rated speed has to be determined. In steady state, the armature voltage is given by

$$V_a = R_a I_{ar} + K_b \omega_{mr} + V_{br}$$

where I_{ar} is the rated armature current, given as 2650 A, ω_{mr} is the rated speed in rad/sec, and V_{br} is the voltage drop across the brushes in the armature circuit and is equal to 2V (given in the problem). To solve this equation, the emf constant has to be solved for from the available data. Recalling that the torque and emf constants are equal, the torque constant can be computed from the rated electromagnetic torque and the rated current as

$$K_t = \frac{T_{er}}{I_{ar}} = \frac{T_s + T_f}{I_{ar}}$$

where the rated electromagnetic torque generated in the machine, T_{er} , is the sum of the rated shaft torque T_s and friction torque T_f . The rated shaft or output torque is obtained from the output power and rated speed as follows:

$$\text{Rated speed, } \omega_{mr} = \frac{2\pi * 600}{60} = 62.83 \text{ rad/sec}$$

$$\text{Rated shaft torque, } T_s = \frac{P_m}{\omega_{mr}} = \frac{1500 * 10^3}{62.83} = 23,873 \text{ N}\cdot\text{m}$$

$$\text{Friction torque, } T_f = B_1 \omega_{mr} = 15 * 62.83 = 942.45 \text{ N}\cdot\text{m}$$

$$\text{The electromagnetic torque, } T_{er} = T_s + T_f = 23,873 + 942.45 = 24,815.45$$

Therefore, the torque constant is

$$K_t = \frac{T_{er}}{I_{ar}} = \frac{24,815.45}{2650} = 9.364 \text{ N}\cdot\text{m/A}$$

$$K_b = 9.364 \text{ V/(rad/sec)}$$

Hence the input armature voltage is computed as

$$V_a = 0.003645 * 2650 + 9.364 * 62.83 + 2 = 600 \text{ V}$$

$$\text{Armature and field power inputs} = V_a I_{ar} + \text{Field power input} = 600 * 2650 + 50,000 = 1640 \text{ kW}$$

$$\text{Output power, } P_m = 1500 \text{ kW}$$

$$\text{Efficiency, } \eta = \frac{P_m}{P_i} = \frac{1500 * 10^3}{1640 * 10^3} * 100 = 91.46 \%$$

Example 2.3

A separately-excited dc motor with the following parameters: $R_a = 0.5 \Omega$, $L_a = 0.003\text{H}$, and $K_b = 0.8 \text{ V/rad/sec}$, is driving a load of $J = 0.0167 \text{ kg}\cdot\text{m}^2$, $B_1 = 0.01 \text{ N}\cdot\text{m/rad/sec}$ with a load torque of $100 \text{ N}\cdot\text{m}$. Its armature is connected to a dc supply voltage of 220 V and is given the rated field current. Find the speed of the motor.

Solution The electromagnetic torque balance is given by

$$T_e = T_l + B_1 \omega_m + J \frac{d\omega_m}{dt}$$

$$\text{In steady state, } \frac{d\omega_m}{dt} = 0$$

$$T_e = T_l + B_1 \omega_m = 100 + 0.01 \omega_m$$

$$T_e = K_b i_a = 100 + 0.01 \omega_m$$

$$i_a = \frac{(100 + 0.01 \omega_m)}{K_b} = (125 + 0.0125 \omega_m)$$

$$e = V - R_a i_a = 220 - 0.5 * (125 + 0.0125 \omega_m) = 157.5 - 0.00625 \omega_m = K_b \omega_m$$

Rearranging in terms of ω_m ,

$$\omega_m(0.8 + 0.00625) = 157.5$$

$$\text{Hence } \omega_m = \frac{157.5}{0.80625} = 195.35 \text{ rad/sec}$$

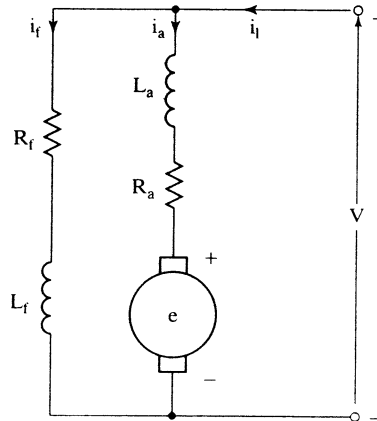


Figure 2.5 dc shunt machine

2.7.2 Shunt-Excited dc Machine

If the field winding is connected in parallel to the armature winding, then the machine goes by the name of *shunt-excited dc machine* or simply by *dc shunt machine*. The equivalent circuit of the machine is shown in Figure 2.5. Note that, in this machine, the field winding does not need a separate power supply, as it does in the case of the separately-excited dc machine. For a constant input voltage, the field current and hence the field flux are constants in this machine. While it is good for a constant-input-voltage operation, it has troubling consequences for variable-voltage operation. In variable-input voltage operation, an independent control of armature and field currents is lost, leading to a coupling of the flux and torque production channels in the machine. This is in contrast to the control simplicity of the separately-excited dc machine. For a fixed dc input voltage, the electromagnetic torque vs. speed characteristic of the dc shunt machine is shown in Figure 2.6. As torque is increased, the armature current increases, and hence the armature voltage drop also increases, while, at the same time, the induced emf is decreased. The reduction in the induced emf is reflected in a lower speed, since the field current is constant in the machine. The drop in speed from its no-load speed is relatively small, and, because of this, the dc shunt machine is considered a constant-speed machine. Such a feature makes it unsuitable for variable-speed application.

2.7.3 Series-Excited dc Machine

If the field winding is connected in series with the armature winding, then it is known as the *series-excited dc machine* or *dc series machine*, and its equivalent circuit is shown in Figure 2.7. It has the same disadvantage as the shunt machine, in that there is no independence between the control of the field and the armature

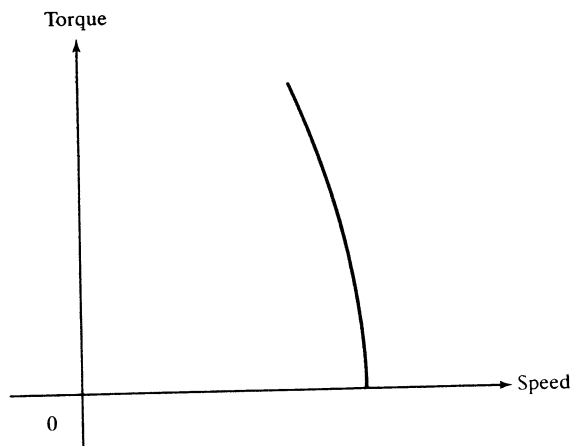


Figure 2.6 Torque-speed characteristics of a dc shunt motor

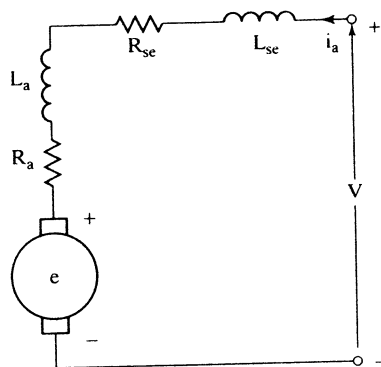


Figure 2.7 dc series machine

currents. The electromagnetic torque of the machine is proportional to the square of the armature current, because the field current is equal to the armature current. At low speeds, a high armature current is feasible, with a large difference between a fixed applied voltage and a small induced emf. This results in high torque at starting and low speeds, making it an ideal choice for applications requiring high starting torques, such as in propulsion. The torque-vs.-speed characteristic for the dc series machine for a fixed dc input voltage is shown in Figure 2.8. With the dependence of the torque on the square of the armature current and the fact that the armature current availability goes down with increasing speed, torque-vs.-speed characteristic resembles a hyperbola. Note that, at zero speed and low speeds, the torque is large but somewhat curtailed from the square current law because of the saturation of the flux path with high currents.

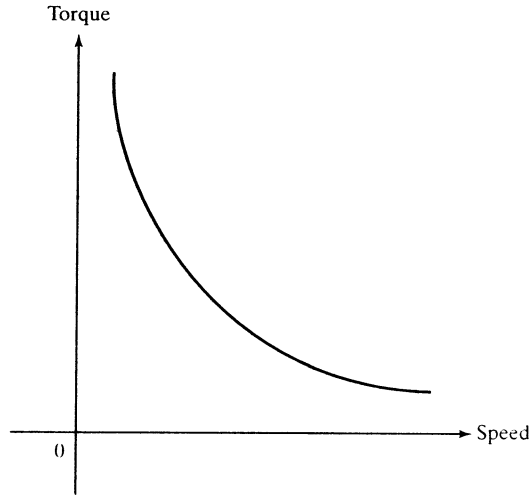


Figure 2.8 Torque-speed characteristic of a dc series motor

Example 2.4

A series-excited dc machine designed for a variable-speed application has the following name-plate details and parameters:

3 hp, 230 V, 2000 rpm

$R_a = 1.5 \, \Omega$, $R_{se} = 0.7 \, \Omega$, $L_a = 0.12 \, \text{H}$, $L_{se} = 0.03 \, \text{H}$, $M = 0.0675 \, \text{H}$, $B_1 = 0.0025 \, \text{N} \cdot \text{m}/(\text{rad}/\text{sec})$

Calculate (i) the input voltage required in steady state to deliver rated torque at rated speed and (ii) the efficiency at this operating point. Assume that a variable voltage source is available for this machine.

Solution (i) The name-plate details give the rated speed and rated power output of the machine, from which the rated torque is evaluated as follows:

$$\text{Rated speed, } \omega_{mr} = \frac{2\pi N_r}{60} = \frac{2\pi * 2000}{60} = 209.52 \, \text{rad/sec}$$

$$\text{Rated output torque, } T_s = \frac{P_m}{\omega_{mr}} = \frac{3 * 745.6}{209.52} = 10.675 \, \text{N} \cdot \text{m}$$

$$\text{Friction torque of the machine, } T_f = B_1 \omega_{mr} = 0.0025 * 209.52 = 0.52 \, \text{N} \cdot \text{m}$$

$$\text{Air gap torque, } T_e = T_s + T_f = 10.675 + 0.52 = 11.195 \, \text{N} \cdot \text{m}$$

The voltage equation of the dc series machine from the equivalent circuit is derived as

$$v = R_a i_a + R_{se} i_f + M i_f \omega_m + L_a \frac{di_a}{dt} + L_{se} \frac{di_f}{dt}$$

where the armature and field current are equal to one another in the series dc machine ($i_f = i_a$) and, in steady state, the derivatives of the currents are zero, resulting in the following expression:

$$V = (R_a + R_{se} + M \omega_m) I_a$$

and air gap torque is given by

$$T_e = M i_a i_a = M i_a^2 = M I_a^2 \quad (\text{N}\cdot\text{m})$$

The air gap torque is computed as 11.195 N·m, and the steady-state armature current is found from the expression above as

$$I_a = \sqrt{\frac{T_e}{M}} = \sqrt{\frac{11.195}{0.0675}} = 12.88 \text{ A}$$

which, upon substitution in the steady-state input voltage equation at the rated speed, gives

$$V = (1.5 + 0.7 + 0.0675 * 209.52) 12.88 = 210.46 \text{ V}$$

- (ii) The input power is $P_i = V I_a = 210.46 * 12.88 = 2,710.45 \text{ W}$
 The output power is $P_m = 3 * 745.6 = 2236.8 \text{ W}$
 Efficiency is $\eta = \frac{P_m}{P_i} = \frac{2236.8}{2710.45} * 100 = 82.5\%$
-

2.7.4 DC Compound Machine

Combining the best features of the series and shunt dc machines by having both a series and shunt field in a machine leads to the *dc compound machine* configuration shown in Figure 2.9. The manner in which the shunt-field winding is connected in relation to the armature and series field provides two kinds of compound dc machine. If the shunt field encompasses the series field and armature windings, then that configuration is known as a *long-shunt compound dc machine*. The *short-shunt compound dc machine* has the shunt field in parallel to the armature winding. In the latter configuration, the shunt-field excitation is a slave to the induced emf and hence the rotor speed, provided that the voltage drop across the armature resistance is negligible compared to the induced emf. Whether the field fluxes of the shunt and series field are opposing or strengthening each other gives two other configurations, known as *differentially* and *cumulatively* compounded dc machines, respectively, for each of the long and short shunt connections.

2.7.5 Permanent-Magnet dc Machine

Instead of an electromagnet with an external dc supply, the excitation can be provided by permanent magnets such as ceramic, alnico, and rare earth varieties: samarium–cobalt and boron–iron–neodymium magnets. The advantage of such an excitation consists in the compactness of the field structure and elimination of resistive losses in the field winding. These features contribute to a compact and cool machine, desirable features for a high-performance motor. The cross section of a permanent-magnet dc machine is shown in Figure 2.10. Note that the armature is similar to other dc motors in construction and performance.

To analyze a dc motor, the constants R_a , L_a , and K_b and the resistance and inductance of field windings are required. Some of them are given in the manufacturer's data sheet. In case of nonavailability of the data, it is helpful to have knowledge of procedures to measure these constants. The next section deals with the measurement of motor parameters.

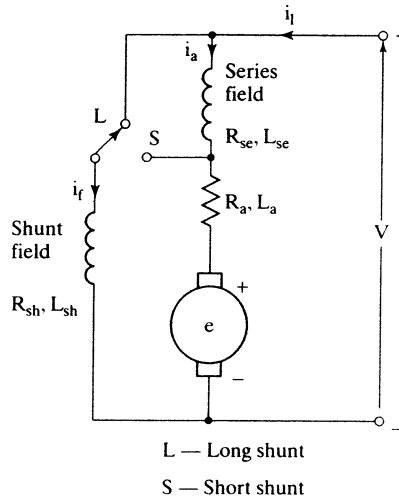


Figure 2.9 DC compound machine

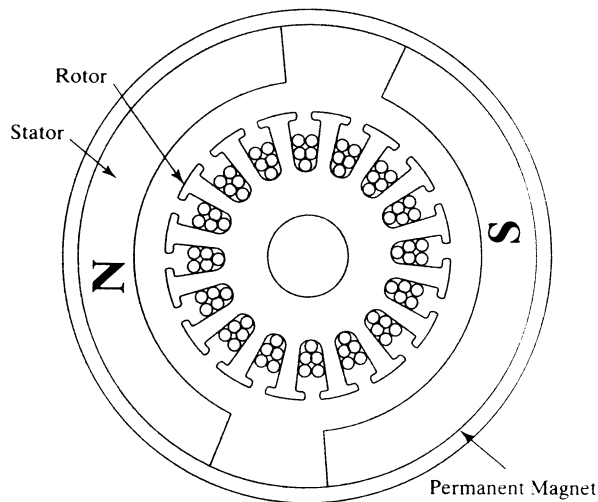


Figure 2.10 Cutaway view of a conventional permanent-magnet dc motor assembly

2.8 MEASUREMENT OF MOTOR CONSTANTS

The following test methods apply to a separately excited dc motor; this type is the most widely used motor for variable-speed applications.

2.8.1 Armature Resistance

The dc value of the armature resistance is measured between the armature terminals by applying a dc voltage to circulate rated armature current. Care should be

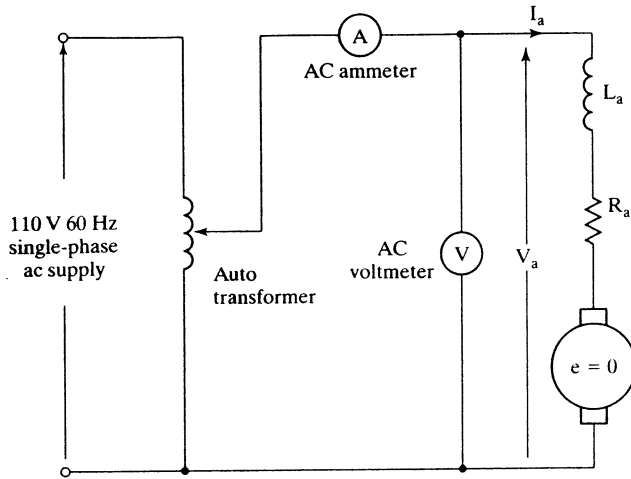


Figure 2.11 Measurement of armature inductance

taken to subtract the brush and contact resistance from the measurement and to correct for the temperature at which the motor is expected to operate at steady state.

2.8.2 Armature Inductance

By applying a low ac voltage through a variac to the armature terminals, the current is measured. The motor has to be at a standstill, keeping the induced emf at zero. Preferably, the residual voltage in the machine is wiped out by repetitive application of positive and negative dc voltage to the armature terminals. The test schematic is shown in Figure 2.11. The inductance is

$$L_a = \frac{\sqrt{\frac{V_a^2}{I_a^2} - R_a^2}}{2\pi f_s} \quad (2.28)$$

where f_s is the frequency in Hz and the armature resistance has to be the ac resistance of the armature winding. Note that this is different from the dc resistance, because skin effect produced by the alternating current.

2.8.3 EMF Constant

Specified field voltage (called rated voltage) is applied and kept constant, and the shaft is rotated by a prime mover (another dc motor) up to the speed given in the name plate (called rated speed or base speed). The armature is open-circuited, with